



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Programme: RDS (Group: 4)

Assignment

BACS2163 SOFTWARE ENGINEERING

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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted works are all our own work and are in our own words.

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1.0 Organisation Background

Tutor Tutor Cemerlang (TTC) Education Center is a tuition centre that was established in the year of 2000 by Dr Ng Hien Ghim. It has a total of 17 branches in Selangor and Kuala Lumpur. They offered several popular subjects to their students such as Malay Language, English Language, Chinese Language, Geography, Science, etc. The tuition centre gains significant prominence in the year of 2020 amid the COVID-19 pandemic by adapting swiftly to the new learning paradigms, such as online and hybrid tuition solutions.

The main purpose of the tuition centre is not only to provide high quality education to its students, but also to bring joy into learning among teenagers. Moreover, they believe that a strong educational basis is the key to achieve success in studies. In order to achieve this goal, a number of available professional and dedicated tutors are hired and well equipped with the latest technology in teaching the students through physical, online and hybrid classes. They are also very experienced and creative in finding new ways to teach every student that they have encountered. For example, They provide exam strategies and answering techniques by referring and researching past examinations and future insight of some key figures from the education system in order to provide their students with the best solutions to answer the exam questions effectively. Another point to be made is that TTC Education Centre is very creative in their ways of teaching their students. To illustrate some examples, the centre distributes colourful and exciting notes to their students in order to attract their attention so that they will not feel bored during their studies. The centre also produces recorded videos to students in order to allow them to revise their class during their free time after their class has ended and also for students who are absent the class can study by watching the recorded video to make sure after coming back to the class they can follow the lesson. With all that being said, this is how the TTC Education has achieved a significant milestone with most of their students completing their exams with flying colours. In the year 2022, the centre has 658 students who achieve above 8A's in Malaysian Certification of Education (SPM). The tuition centre also made the tuition fee affordable for everyone to study in a suitable environment. It costs from RM 66 to RM 95 per subject for secondary students and from RM 41 to RM70 for primary students depending on the subjects that are taken for each month. Moreover, they also offer free seminars for students to ensure that they are able to cope with the main key points of every chapter of the particular subject.

Due to the large number of students attending the tuition centre, they have to come up with a student information system to help them sort out all the information such as students' information, schedules, grades and so on in order to keep track of every student's progress more efficiently. Additionally, the student information system also includes the finance management, whose

responsibility is to keep track of the finances of the centre. One of the main reasons for this management to be formed is to be able to remind the student or parent to pay the tuition fees. There will also be an attendance system in order for parents to keep track of their child's attendance and performances. The system also can help to improve communication between the students and parents. It can share the important updated information and reminders through the system.

Objective: Develop a student information system in order to manage student details such as grades as well as improve communication between tuition centres with students and their parents.

Cost: RM 7,000

Schedule: 4.5 months per incremental or module

2.0 Requirement Gathering Techniques

Requirement gathering is a vital component in developing a system. There are multiple techniques that can be used to gather all the necessary requirements efficiently. In this case, we will be using interviews and questionnaires in order to gather the information and requirements.

2.1 Interview

An interview is a structured conversation involving either a group of people or an one-on-one interaction. In this case, interviews between the software developers, the representative of Tutor Tutor Cemerlang (TTC) Education Center and its stakeholders will be conducted in order to collect and gather the requirements for the software. Structured questions that are prepared thoroughly will be the key in leading the discussion to gain specific functional requirements and also the improvements that are needed to be made on the system. These questions will be given to the representatives and the stakeholders of Tutor Tutor Cemerlang (TTC) Education Center such as the owner, administrator, suppliers, tutors and the student's parents. Through the interviews, the representatives and the stakeholders will be more open to express their views and suggestions to the system developers to gather the requirements for the system.

One of the key figures that should be mentioned first is the owner of Tutor Tutor Cemerlang (TTC) Education Center. The owner is the one who will be the one giving the direction in shaping the tuition centre itself. Therefore, the owner will be the first priority to be interviewed due to the fact that the requirements given by the owner will match with the company's goal consistently. Moreover, the staff such as the administrator and the tutors are prioritised as they are the core members that drive the operations and the success of the tuition centre. The information gained from them will be invaluable due to their direct engagement with students and parents who are the primary stakeholders of the tuition centre. Suppliers of the tuition centre who provide teaching materials and also long-term key collaborators of the tuition centre will also be interviewed in order to provide additional functional requirements which will help ease their operations with the tuition centre. The parties mentioned above will help us guarantee that the final product will satisfy their expectation all together.

Interview is chosen as a technique to gather the requirements due to its effectiveness in facilitating direct communication between the development team and the education centre. This technique also helps in exploring complex topics and clarifying ambiguities. Moreover, the most important part about interviews is that it provides opportunities to build rapport with the participants of the interview, fostering an open environment where everyone is comfortable in sharing their insight and

perspectives which will contribute to the cause of gathering the requirements and in the end lead to the success of the project all in all.

2.1.0 Interview Plan

Interviewer : System Analyst

Interviewee : Top management

Objectives :

- Understand the current existing system
- Determines the problem that has been encountered in the current system
- Establish users requirements or needs in the new student information system

Duration : 15 minutes per sessions

Sampling : 8 individuals who has used the current student information system (Top management or staff)

Questions :

1. What is your role in using the current system?
2. Are you satisfied with the current student information system? Why?
3. What are the key objectives that you wish to achieve with the new student information system ?
4. What are the limitations you see in the current student information system?
5. What are the specific functions that you think are crucial for the new student information system ?
6. Have you received any complaints from customers while using the current system? If so, what type of complaints?
7. What future growth or scalability needs should we consider in the system design?

2.1.1 Requirements Gathered

After conducting interviews with the TTC's staff members, we gain valuable insights and information about the changes that should be made in their old system. TTC brings up the need to implement an online system to help automate key processes of the tuition centre such as recording and tracking students' performance and attendance. This process is done manually in the current system which is prone to errors and mistakes. In addition, the teachers are concerned about the fact that manually entering data, such as attendance for each student one at a time, is time-consuming and inefficient. Moreover, they mentioned that errors such as incorrect contact information of the student's parent or guardian will occur and it will pose a big risk if the situation is dire. Another issue that was brought up during the interview is the lack of online payment services of their current system. This is a major inconvenience because in this day and age, most parents and guardians of the students prefer the security and convenience of digital payment options.

2.2 Questionnaire

Questionnaire is another method that is no less important in the requirement gathering process. A questionnaire is a series of questions that are designed specifically to facilitate the collecting of information for research, evaluation or decision-making. For this case, questions will be crafted using an online platform such as Google Form to help collect the data more efficiently. The questionnaire will have both open-ended and closed-ended multiple choice questions depending on the respective questions. The respondents, who are the representatives and stakeholders of Tutor Tutor Cemerlang (TTC) Education Center will receive an email that has the questionnaire attached to it and they can answer it at any time anywhere.

The owner of Tutor Tutor Cemerlang (TTC) Education Center can help arrange the distribution of the questionnaires which are specifically crafted to cater with different groups of individuals within the tuition centre. The administrators will be answering the questionnaire to provide their views on the user interface design that satisfies them. The tutors can help distribute the questionnaire to students who are at the age of 12 years old or above due to the fact that these students have already witnessed the current advancement of technology around us. Their response will provide us valuable insight on the user experience aspect from a student perspective.

By using questionnaires as one of our requirements gathering techniques, we are able to easily distribute them to all related groups of individuals in a very fast and efficient way. This would allow us to save a lot of time during this phase of the process and be prepared for our next move without any delay. Furthermore, the whole process of crafting, distributing and analysing the questionnaire itself is inexpensive, making it one of the most cost-effective techniques in comparison. In addition, online platforms like Google Form are available for anyone to use at any time with just a click of a button. Some other platforms also provide many ways such as using a bar chart, pie chart and so on to present the data that was collected. Last but not least, the questionnaire is anonymous. Respondents will be more likely to provide more honest feedback without having their names be recorded.

2.2.0 Questionnaire Plan

Target : Administrators, students and parents

Distribution Periods : 11 July 2024 - 17 July 2024

Sampling : 250

Mode : Softcopy

Objectives :

- Gather insights from respondents regarding their experiences and challenges with current student information system
- Identify specific preferences and needs for functionalities and features for the new online student information system
- Fully understand the expectations and concerns of all users groups in regard to the system efficiency, usability and accessibility

Instructions :

1. Fill out the form as truthfully as possible (*Compulsory field)
 2. To enter your answers, click on the text area and start typing.
 3. To check off choices on the form (checkbox), click in the boxed area and a "✓" should appear.
 4. Please answer each question.
 5. To send the completed form successfully, click on the "Submit Form" button at the bottom of the page.
-

Questions :

Administrators

1. Do you find it challenging to manually manage student performance and attendance records? What are the challenges you faced?
2. What features would you find most helpful and fitting in automating student performance and attendance tracking for the new online student information system?
3. Do you have encountered any issues when contacting the students or their parents/guardians with the given contact information which is recorded manually? If yes, please explain.
4. How do you manage the tuition payment process? Do you think an online payment system can be beneficial for your day to day operation? Why or why not?

5. Are there any other tasks related to student management that you deem as a time-consuming process or prone to errors in the current system?
6. What are the specific features that you would expect for the new system to have to improve your workflow?

Students

1. How do you currently check your own performance and attendance records? Is it easy to access?
2. Do you have any suggestions on features that could make it easier for you to track your progress and attendance online?
3. Are there any aspects of the current system that you find inefficient and frustrating?

Parents / Guardians

1. How do you currently receive updates on your child's performance and attendance? Is it convenient for you?
2. Do you prefer an online system for tracking your child's academic progress and attendance? Why or why not?
3. How do you make tuition payments currently? Would you prefer a secure online payment option?
4. Are there any other features that you would expect to see in an online system to improve convenience for parents?

2.1.1 Requirements Gathered

After distributing the questionnaires to teachers, parents and guardians of the students, new additional requirements have been suggested by them for the new system. TTC requires a system that can help compile and generate reports on the students' examination results for easy access by students, parents and teachers. Moreover, the responses of the questionnaires indicated the need for the system to include features that could inform students and parents about the student's timetable schedules and also other announcements.

3.0 Problems of Existing System

3.1 Communication Issue

Effective communication is vital in an educational setting, as it helps to ensure all parties are well-informed and engaged. In the current existing system, when announcements such as homework or mock exam notices need to be placed manually on the notice boards or sent by the admin through WhatsApp groups, the high number of students may lead to some of them being confused on which announcements are made for them in particular. Information such as the marks of the examination or the payment fee for should be informed individually and respectively to the parents. The current system cannot perform such tasks as it is proven to be quite taxing for the administrator to manage. This will also cause a delay in informing everyone on any particular matter that is happening around them. Without this feature, it will be very hard to have the parents to cooperate accordingly.

3.2 Data Management Issue

In the current system, manual data entry by administrators is prone to errors, which can result in incorrect information being recorded within the system, posing serious consequences, especially in emergency situations when accurate student contact details are critical. Additionally, the duplication of student records due to different administrators separately entering the same student's information can lead to confusion, inefficiencies, and inaccuracies. Data inconsistency further exacerbates these issues, as incomplete information and varying data formats such as different administrators using various formats for dates and names can cause significant confusion and errors in processing.

3.3 Security Vulnerabilities

The unauthorised access to the student information system can lead to the exposure of sensitive data, including personal information, academic records, and financial information. The current system, being offline, is exposed to everyone and it is easily accessible, making it vulnerable to physical breaches and unauthorised access. This can result in data theft, identity fraud, and other malicious activities, which in the end, compromise the trust and safety of students, parents, and staff.

3.4 Manual Student Information Management

Handling student information manually can lead to significant challenges and risks. When records are maintained on paper or through non-integrated digital systems, there's a higher likelihood of losing or misplacing important documents. Manual record-keeping is also prone to human error, such as misentries or omissions, which can lead to inaccuracies or loss in student information. Furthermore, the lack of centralised data management makes it difficult to update and track all the changes consistently, leading to potential gaps in information. This can cause a lot of issues in ensuring that all records are constantly updated and accurate, affecting everything from academic performance tracking to communication with parents. The inefficiency of manual handling also means that retrieving or cross-referencing information can be proven to be time-consuming, adding to the administrative burden and potentially leading to delays in addressing students' needs.

3.5 Fee Billing Issue

Parents are required to visit the tuition centre in person to pay fees, which is said to be inconvenient, especially for those with tight schedules who struggle to find a balance between work, household responsibilities, and other commitments. Some students, especially the primary students, are not capable enough in bringing money to the centre on their own to pay the fee due to safety concerns and the risk of the money being lost, stolen, or misused. Additionally, the manual process of checking payments and generating invoices can result in long queues and waiting periods during the end of the month when the due date approaches. Many parents would prefer the flexibility of online payment methods that allow transactions to be completed at their own convenience, anytime and anywhere.

4.0 Software Quality Attributes

Software Quality Attributes is a set of criteria that indicates the objectives and success of a software. We have selected 5 software quality attributes in order to succeed in the implementation of the new student information system. These software quality attributes are efficient, reliable, secure, usable and flexible .

4.1 Efficient

A student information system must always be efficient in its functions and features so that users can complete their task in a short amount of time. For example, the system should be able to update the timetable and schedules of the students' classes within only a few seconds when a staff member makes changes to it. This will prevent situations like short notice from happening. Another example would be when a student or a parent who needs to view their timetable, take attendance or make a payment, there should not be any delay or a waiting page in doing so. With that being said, making sure that efficiency is one of the quality attributes of the project would contribute to the users' amount of time not being wasted.

4.2 Reliable

A reliable student information system is essential because it ensures the accuracy and dependability of critical data such as student attendance records, academic grades, class schedules, and other pertinent information like contact details without having to worry about any failure due to the hardware or software. For a tuition centre administrator, a dependable system ensures accurate tracking of student attendance, academic performance, and class schedules, thereby eliminating mistakes that could cause issues with processes. Teachers rely on the system to view the latest information, which is crucial for evaluating, planning lessons, and tracking students' progress. Students benefit from consistent access to their performance indicators through a reliable system, helping them to stay aware of their achievements and the areas they should focus on improvement. Parents use the system to communicate with the educational institution and receive real-time updates on their child's progress. This reliability is significant because it enables users to make informed decisions for future use, such as monitoring academic progress, creating personalised study plans, and ensuring timely communication. Consequently, by providing easy access to trustworthy information, a dependable student information system not only enhances operational efficiency but also strengthens the overall learning experience.

4.3 Secure

A software system is vital, as it should always be secured and protect sensitive information and details from any unauthorised access. The new student information system and its data must be safeguarded from being threatened or attacked by unauthorised users. The data must be protected at all cost from being compromised or leaked to unapproved third parties. If a system user, either a staff member or a student, wants to use the new student information system, they must use their own authorised account to log in using their student identification number and password.

Furthermore, only a few selected accounts have specific access to some specific features and some of them do not. For example, accounts such as the student accounts, are unable to access sensitive data and amend any of the functions in the system . Sensitive data, such as staff information or learning materials, should be encrypted with a special password to protect the privacy of staff details and copyrights for the learning materials. Therefore, It should only be able to be opened by authorised users such as the administration staff that are allowed to have access to the system's special features. In addition, authentication mechanisms such as face recognition and fingerprints should be implemented in order to ensure the system is secured and reliable for users with the appropriate authentication. Thus, the new student information system will be trustworthy, secure and safe.

4.4 Usable

The new student information system should be easy to use and provide a satisfying experience to the users. Firstly, we should always ensure that every function in the student information system is able to be accessed and function well. User friendly features should always be taken as consideration while we are designing the new system interfaces. This allows users to learn how to use our new student information system effortlessly without any additional training or guidance. Besides that, we should also build the system with ease of data entries. For instance, we should include an example in each input box on the student registration page. The new student information system should include a data validation check in order to minimise the potential of entering wrong information. It should also always prompt an error message to the users when the users commit a mistake such as entering the wrong format of data entries or typo errors. By having a prompt message after errors have been made, users can quickly and conveniently identify the mistakes that have been made and amend it after. Hence, users are able to perform tasks quickly and efficiently after they are familiar with the new student information system.

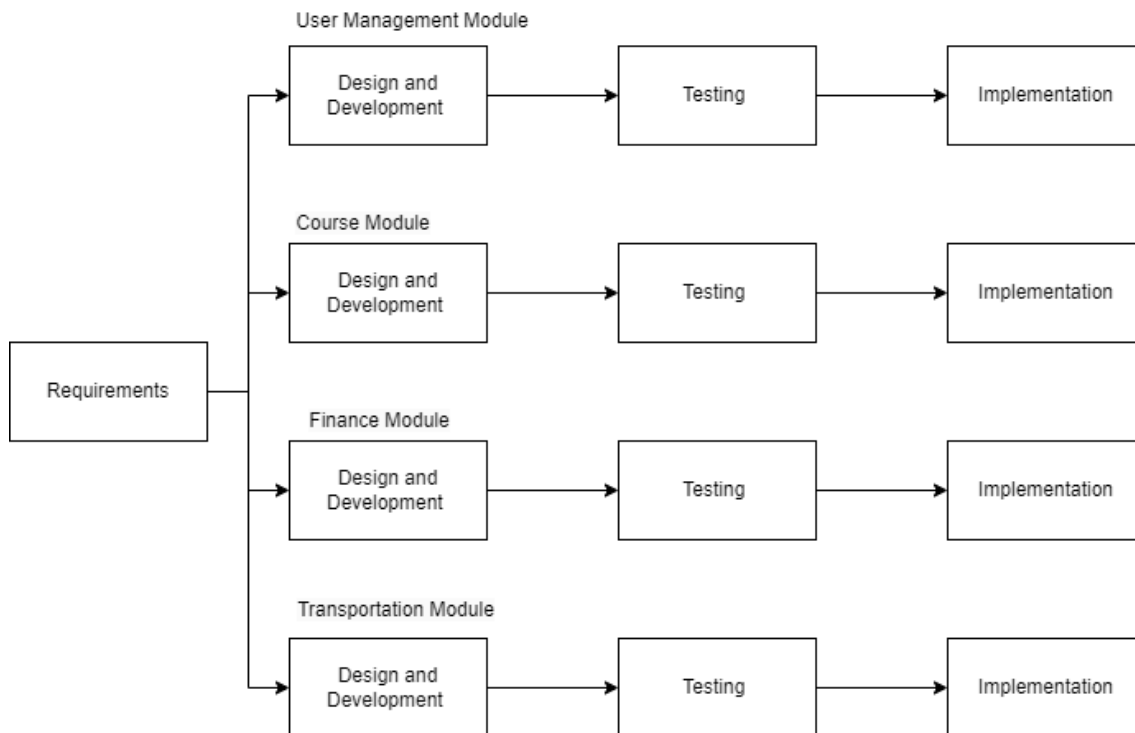
4.5 Flexible

A flexible student information system is important as it is the ability of the software systems to adapt to various situations, environments, and circumstances without requiring significant changes to be made to its software structures. The new student information system should be able to be modified for new purposes, operated in different countries or transferred to a new platform without the major changes of the software. A flexible system should prevent redundancy in the database, even when entering data from a different device or account. For example, the student information system can be used by all the students and staff using different types of devices such as laptop, desktop, tablet or smartphone. It should provide the same output on any of the devices. This system must be able to be operated in different operating systems such as Windows, Android, IOS, MacOS, etc. It is essential as it guarantees that students or staff that are using devices from other manufacturers or different operating systems, can access their academic information and resources.

5.0 Software Process Model

5.1 Incremental Model

Incremental model is a type of software process model that is conducted by dividing the model into a multiple standalone module or stages in the software development cycle. Every module in the incremental model has four phases, which are the requirements phase, design phase, implementation phase and the testing phases.



5.1.0 Requirement Phase

The requirements phase in the incremental model involves gathering and analysing the requirements and the details of the project, which includes the objectives, scopes, functions and constraints. It also includes defining the overall project plan and identifying the resources needed in order to complete the project. In the incremental model, clear requirements are needed to be gathered and understood by the development team in order to minimise the mistakes that might occur during the system development phase while also maintaining the satisfaction of the clients. Once the requirements and the resources have been confirmed and documented, it will undergo a sort of review and revisions before proceeding to the next phase which is the design phase.

5.1.1 Design and Development Phase

During this phase, the project team will start to plan and design the system in detail. The design of the system and the technologies that are required to implement the system will be discussed thoroughly and chosen in the design phase in order to meet the users specific requirements.

5.1.2 Testing Phase

The testing phase is conducted right after the design and development phase and its purpose is to ensure the developed solutions meet the requirements and functions as expected. During this phase, any sort of defects, bugs, or issues that are identified can be addressed to the development team to fix it before implementation. Any possible changes that are needed to be made can also be done before the implementation phase in order to ensure that each increment is stable, reliable, and ready for deployment, ultimately leading to a more robust and high-quality system upon full implementation.

5.1.3 Implementation Phase

This phase takes over right after the testing phase where the tested software increments will be taken from the testing environment to a live production environment. End-users will have access to it and are able to use it to its full content to ensure that the functionality of the increments is as intended.

5.2 Justification on Using Incremental Model

5.2.0 Clear Requirements

The incremental model requires that the requirements are gathered and documented clearly. This is because clear requirements can help to ensure that the development team has a clear understanding and the knowledge needed to build the new student information system. Since the student information system has a fixed requisite compared to other complex systems. Therefore, the basics of the system can be planned and discussed without any prior comprehensive research.

5.2.1 Staff Shortage Issue

The incremental model allows the development team to develop a new system while considering shortage of staff. This is because the incremental model is built according to modules or stages, which allows the development team to build stages by stages. This could benefit the project as a whole due to the fact that each of the modules will be concentrated one at a time. Each component will be more refined before moving on to the next.

5.2.2 Flexible

The incremental model can be said to be flexible compared to other software process models. It has a higher acceptance rate when a sudden change of requirements is needed. Since the incremental module is built off module by module, it will be less expensive to change the scope or requirements of the system during the development of the new system. Besides that, the processes of testing and debugging will be easy as the development team needs to undergo testing phases before each stage is implemented. The module has a smaller iteration compared to the other model. Therefore, the development team is able to test and identify the errors or mistakes that have been made efficiently.

5.2.3 Early Core Function Delivery

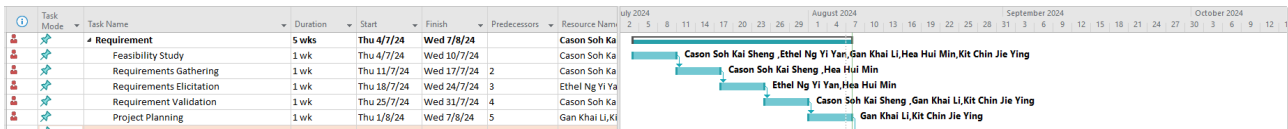
The incremental model is chosen as the software process model for the student information system as core functions are operational before the system is fully developed and implemented. It allows users to utilise the system with some basic functions first, while at the same time updating and releasing new features to the system to provide a better user experience and satisfaction. By releasing the module incrementally, it allows system designers and developers to provide values to the stakeholder earlier. This can also help the system developers to get feedback and make the appropriate adjustments from the new users requirement earlier.

6.0 Project Planning Schedule

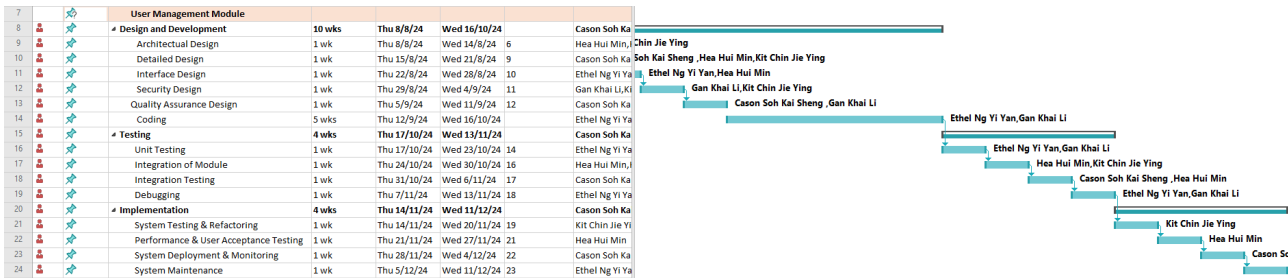
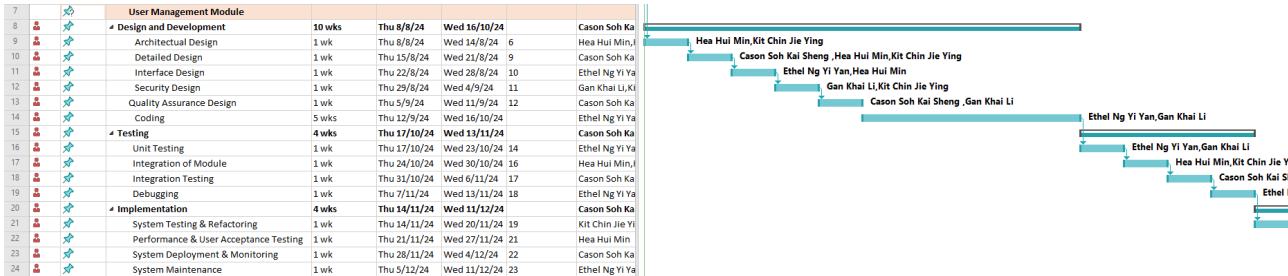
6.1 Task Allocation List

	Task Name	Predecessors	Duration	Overlap
1	Requirements	-	5 weeks	None
	User Management		18 weeks	
2	Design and Development	1	10 weeks	None
3	Testing	2	4 weeks	None
4	Implementation	3	4 weeks	None
	Course Module		18 weeks	
5	Design and Development	4	10 weeks	None
6	Testing	5	4 weeks	None
7	Implementation	6	4 weeks	None
	Finance Module		18 weeks	
8	Design and Development	7	10 weeks	None
9	Testing	8	4 weeks	None
10	Implementation	9	4 weeks	None
	Transportation Module		18 weeks	
11	Design and Development	10	10 weeks	None
12	Testing	11	4 weeks	None
13	Implementation	12	4 weeks	None

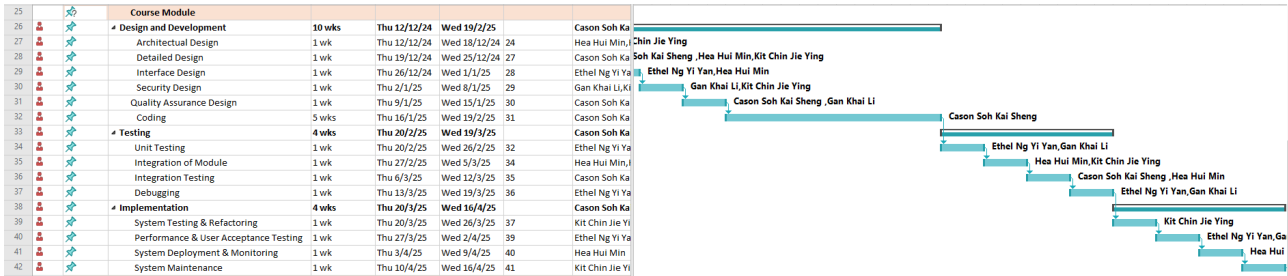
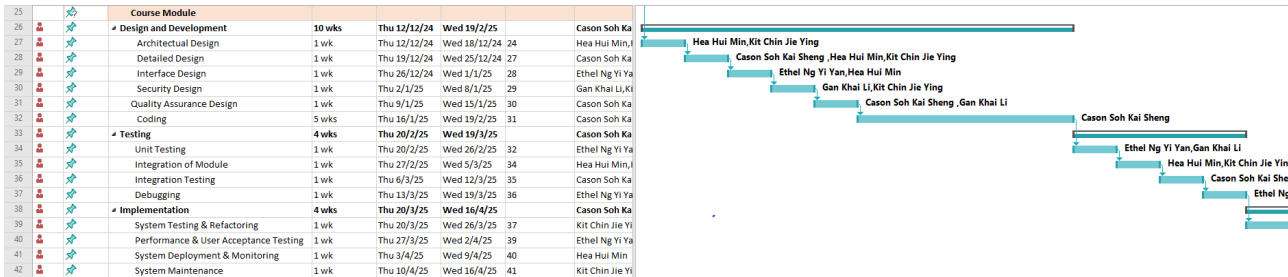
6.2 Gantt Chart



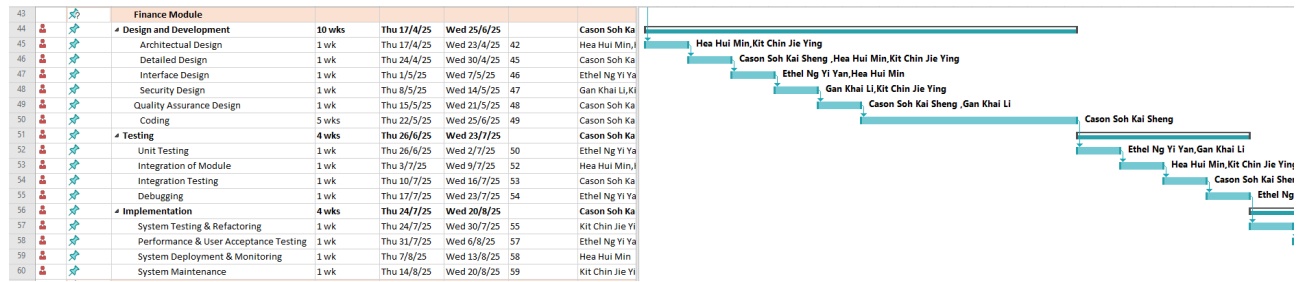
6.2.0 Incremental 1 : User Management Module



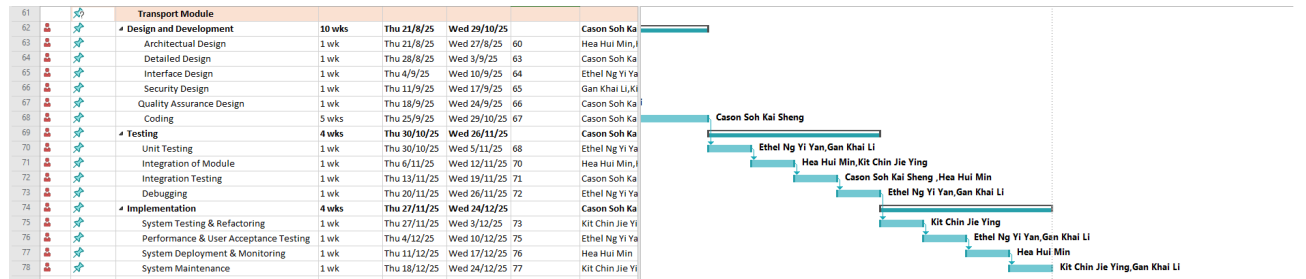
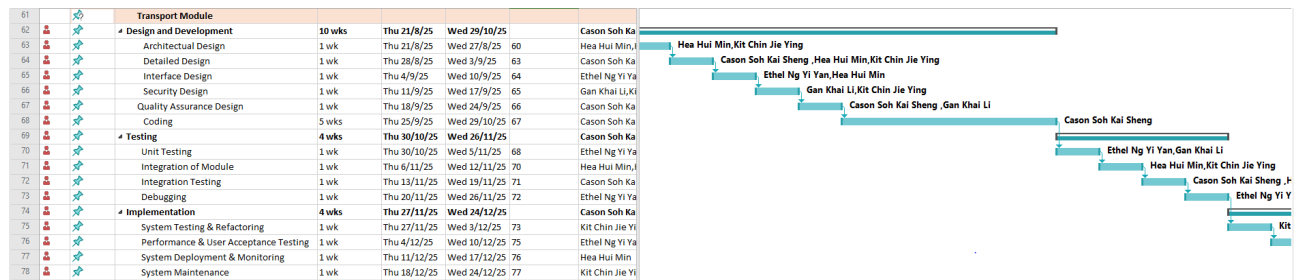
6.2.1 Incremental 2 : Course Module



6.2.2 Incremental 3 : Finance Module



6.2.3 Incremental 4 : Transport Module



7.0 System Requirements

7.1 Functional Requirement

7.1.1 Search (Cason Soh Kai Sheng)

- 1.0 The system shall provide a search bar in a designated area of the application
- 1.1 The system shall allow the users to enter a student's ID in the search box
- 1.2 The system shall display the search result to show relevant information based on the student's ID entered by the user
- 1.3 The system shall display an error message indicating that there is no result that matches with the student's ID
- 1.4 The system shall allow the users to enter a student's name in the search box
- 1.5 The system shall display the search result to show relevant information based on the student's name entered by the user
- 1.6 The system shall display an error message indicating that there is no result that matches with the student's name
- 1.7 The system shall allow the users to enter a course code in the search box
- 1.8 The system shall display the search result to show relevant information based on the course's ID entered by the user
- 1.9 The system shall display an error message indicating that there is no result that matches with the course's ID

7.1.2 Registration (Cason Soh Kai Sheng)

- 1.0 The system shall allow student to enter their personal information such as their full name, email address and phone number
- 1.1 The system shall prompt an error message if the student enters the wrong format of their personal information
- 1.2 The system shall allow student to reenter the user's personal information that was entered in the wrong format
- 1.3 The system shall check for existing accounts using the same personal information such as email address and phone number to prevent duplicate registrations
- 1.4 The system shall generate a unique ID for a new student after the student's personal information has successfully entered
- 1.5 The system shall allow the student to create a new password
- 1.6 The system shall verify if the password is entered according to a certain format(eg. 12 numeric format)

- 1.7 The system shall prompt the user to reenter the password when the password does not meet the required format
- 1.8 The system shall redirect the student to the login page once the registration process is completed

7.1.3 Calculate and generate examination reports for grades(Grade from school exam, Grade from tuition homework/exam) (Hea Hui Min)

- 1.0 The system shall display the examination report of grades
- 1.1 The system shall allow users to search examination report of grades
- 1.2 The system shall allow users to export the examination report of grades
- 1.3 The system shall allow teachers to input the marks of the exam and tuition homework
- 1.4 The system shall calculate the report of grades from exam and tuition homework
- 1.5 The system shall allow users to view the history of previous report of grades
- 1.6 The system shall prompt out the error message for any reports of grades that does not exist
- 1.7 The system shall notify users via email when the report of grades is released
- 1.8 The system shall prompt out the error message for unsuccessfully generating reports of grades

7.1.4 Modify/Delete Existing Record (Hea Hui Min)

- 1.0 The system shall display the examination report of grades
- 1.1 The system shall allow students to modify basic information in their student profile
- 1.2 The system shall allow administrators to modify the student information
- 1.3 The system shall allow teachers to remove the examination report of grades
- 1.4 The system shall allow teachers to modify the examination report of grades
- 1.5 The system shall display the examination report of grades after edited
- 1.6 The system shall allow teachers to check the history of the original examination report of grades
- 1.7 The system shall allow teachers to add the examination report of grades
- 1.8 The system shall allow teacher access the time of post of the examination report of grades
- 1.9 The system shall prompt out the error message when input data is invalid
- 1.10 The system shall prompt out the error message for failed to delete examination report of grades
- 1.11 The system shall prompt out the error message for failed modify examination report of grades

7.1.5 Login (Ethel Ng Yi Yan)

- 1.0 The system shall allow student to enter the ID and password
- 1.1 The system shall verify the user ID and password
- 1.2 The system shall prompt an error message for invalid ID or password
- 1.3 The system shall allows the student to reenter the ID and password
- 1.4 The system shall lock the account if invalid ID and password is entered more than 5 times
- 1.5 The system shall only allow administrator to unlock a locked account
- 1.6 The system shall allow student to reset the password if the password is forgotten
- 1.7 The system shall allow student to enter the user ID and IC number to verify account
- 1.8 The system shall verify their user ID and IC number to reset the password for the account

7.1.6 Attendance Management (Ethel Ng Yi Yan)

- 1.0 The system shall allow student to enter the attendance code
- 1.1 The system shall verify the attendance code
- 1.2 The system shall prompt an error message for invalid attendance code
- 1.3 The system shall allow student to reenter the attendance code
- 1.4 The system shall allow student to generate attendance report
- 1.5 The system shall allow student to view the attendance report
- 1.6 The system shall allow teacher to check student's attendance
- 1.7 The system shall send a warning message to the student who absent to class more than 3 times without a valid reason for each subject
- 1.8 The system shall allow teacher to help students to take attendance
- 1.9 The system shall allow the teacher to edit student's attendance

7.1.7 Timetable Management (Gan Khai Li)

- 1.0 The system shall allow the administrator to create the timetable
- 1.1 The system shall allow the administrator to modify the existing timetable
- 1.2 The system shall allow the administrator to delete the timetable
- 1.3 The system shall allow the user to view the timetable
- 1.4 The system shall allow the administrator to assign teachers to specific classes
- 1.5 The system shall allow the administrator to assign students to specific classes
- 1.6 The system shall allow the administrator to reassign or replace teacher in the timetable
- 1.7 The system shall allow the administrator to create or schedule mock exam session
- 1.8 The system shall send a notification to the teachers and students when the timetable is changed

7.1.8 Real-time Announcement Management (Gan Khai Li)

- 1.0 The system shall allow the administrator to create a new announcement
- 1.1 The system shall allow the administrator to modify an existing announcement
- 1.2 The system shall allow the administrator to delete announcements
- 1.3 The system shall allow the administrator to attach file to announcements
- 1.4 The system shall allow the user to view announcements
- 1.5 The system shall support immediate announcements in emergency situations
- 1.6 The system shall allow the administrator to set the deadline for announcements
- 1.7 The system shall allow announcements to be targeted to specific groups
- 1.8 The system shall track the reading status of announcements for each user
- 1.9 The system shall allow the users to search the announcements

7.1.9 Finance Management (Kit Chin Jie Ying)

- 1.0 The system shall allow administrator to upload payment
- 1.1 The system shall automatically update payment status
- 1.2 The system shall generate invoice to student after complete the payment
- 1.3 The system shall keep the user payment history
- 1.4 The system shall allow administrator to generate the finance report for analytics use
- 1.5 The system shall allow the administrator to view the history of payment in real-time request
- 1.6 The system shall allow the student to view the history of payment in real-time request
- 1.7 The system shall allow administrator to view the payments details
- 1.8 The system shall allow student to view the payments details
- 1.9 The system shall allow student to choose payment methods

7.1.10 Transport Booking (Kit Chin Jie Ying)

- 1.0 The system shall allow student to book transport
- 1.1 The system shall allow student to check booking status
- 1.2 The system shall allow student to check the transport scheduling
- 1.3 The system shall provide student the information on scheduled routes
- 1.4 The system shall provide the detailed information of transport that have been scheduled
- 1.5 The system shall allow student to cancel the booking
- 1.6 The system shall allow administrator to update the transport timetable
- 1.7 The system shall allow student to enter personal details such as name, contact details, address and enter pick up and return time

7.2 Non-Functional Requirement

7.2.1 Product

1. Usability

- The system has a very low learning time, provides user guidance and uses familiar icons.

2. Performance

- The system should load a student's profile or exam schedule within 3 seconds under normal conditions.
- The system should be able to handle more users and data without slowing down.
- The system must process at least 100 transactions per second, even when multiple requests come in at the same time.

3. Storage

- The system should store data efficiently, minimizing space usage and ensuring quick access.
- The system should support scalable storage that can grow as data needs increase without major changes to the system.

4. Reliability

- The system downtime should not be more than 0.05 %
- The system should ensure the data storing is accurate and consistent.
- The system should be regularly automated backup.
- The system should handle errors smoothly, give clear error messages, and let users recover or retry without losing data.

5. Portability

- The system should work on different operating systems without needing major changes
- The system should be compatible with major browsers.

7.2.2 Organisation

- The new system should be developed using the object the Object-Oriented Analysis and Design Development Methodology.
- The new system is implemented based on the Oracle database.

7.2.3 External

1. Privacy

- The system shall not disclose any personal information apart from their name, user ID and payment status to the administration staff.

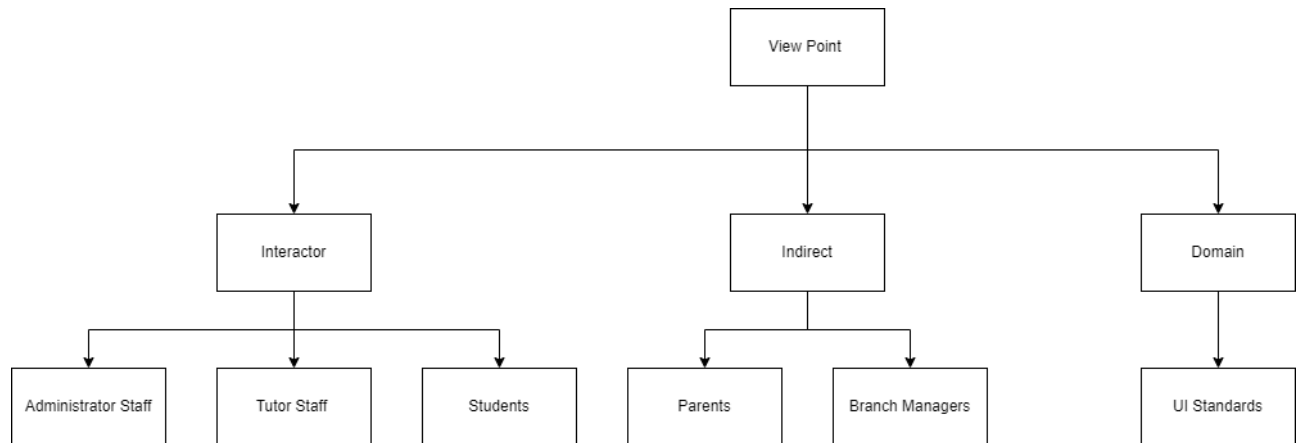
The system shall be responsible to private the student information such as contact number, address and email. This is because users only allow the staff access to view the information for registration and to use this system. It is responsible that the student information is only used for this system only. If the system does not private the student's important information, it will cause a lot of problems to students and students ' families. For instance, if someone gains access to student information, there is a risk that he will pretend to be the student online to access their personal data to login to various websites or commit cyber crimes such as tax fraud.

2. Safety

- The system shall secure the safety of the system and protect the system from unauthorised access.

The system shall be responsible for making sure that the system is safe. For example, the system shall provide the authentication mechanism to verify the identity of users or systems accessing the software. Access controls determine what actions or data each authenticated entity is allowed to access. This action is to protect the system from unauthorized access, to ensure that only our tuition students, teacher and administrator staff can access the system. So if the system is not safe, when a student is doing the payment, it has the risk of being stolen and also cause the student important information such as contact number, address, ic number and email will be leaked.

8.0 Viewpoint-Oriented Analysis



8.1 Administrator Staff (Hea Hui Min)

Administrator staff act as an interactor in the viewpoint-oriented analysis due to the reason that administrator staff is most of the time that directly uses the system. The main purpose of administrator staff is to help to store the student information and preferably in the form of online student profiles. Administrator staff have access to add, modify or remove student's information in the application. For example, when a new student's register for tuition, administrator staff need to upload the student's information to the system. The administrator staff also have responsibility to update the latest announcement to the students such as the timetable change time, due date about to pay tuition fee and teacher cancelled class. Apart from that, administrator staff have access to overseas tuition fees and payments. This is due to the reason that administrator staff need to generate the report about the tuition fees and payments. When student's need to add one more subject, it may also cause the administrator staff to calculate the tuition fees again and update it.

8.2 Tutor Staff (Ethel Ng Yi Yan)

Tutor staff act as an interactor in the viewpoint-oriented analysis, which are the direct users to the student information system. This is because tutor staff use the system in order to view and manage their timetable as well as checking their students progress. A tutor staff usually utilises the student information system directly to enter and update the student's marks after a mock exam is given. Besides that, the tutors are also responsible to supervise the student's attendance status. For example, the tutor staff should help the students to take their attendance if their students forgot to take the attendance during the tuition period. Moreover, the tutor staff is also responsible for scheduling the timetable for their classes especially for the class replacement. As the tutor staff is one of the interactors which will directly use the system, opinions and requirements of a tutor staff is important in order to understand the user requirements and provide a user's satisfaction.

8.3 Students (Kit Chin Jie Ying)

The role of students in using the student information system is essential as students are the primary users with direct access to the system. Their active participation and engagement with the system significantly impact the overall effectiveness of the system. Students use the application to mark their attendance for each class session. This system allows them to keep track of their attendance record. This system also provides a platform for the student to check the latest information or announcements from the tuition centres. Through this, it can ensure that students are notified of any updates, adjustments, or specific instructions relating to their courses. Through the system, students can view their timetable to help them to manage the time effectively by tracking the class timings and dates as there may be any potential changes of the timetable. The system enables students to check their grades and performance on the tests, providing insights into their academic progress. Furthermore, students can select the subjects they wish to study and indicate the amount of time they plan to dedicate to each subject. This ensures they are enrolled in courses that align with their academic goals and manage their study time appropriately. The feedback and comments from the students are crucial as they are the primary users of the system and their experiences directly influence the system's effectiveness and drive improvements, ensuring that it better meets their needs.

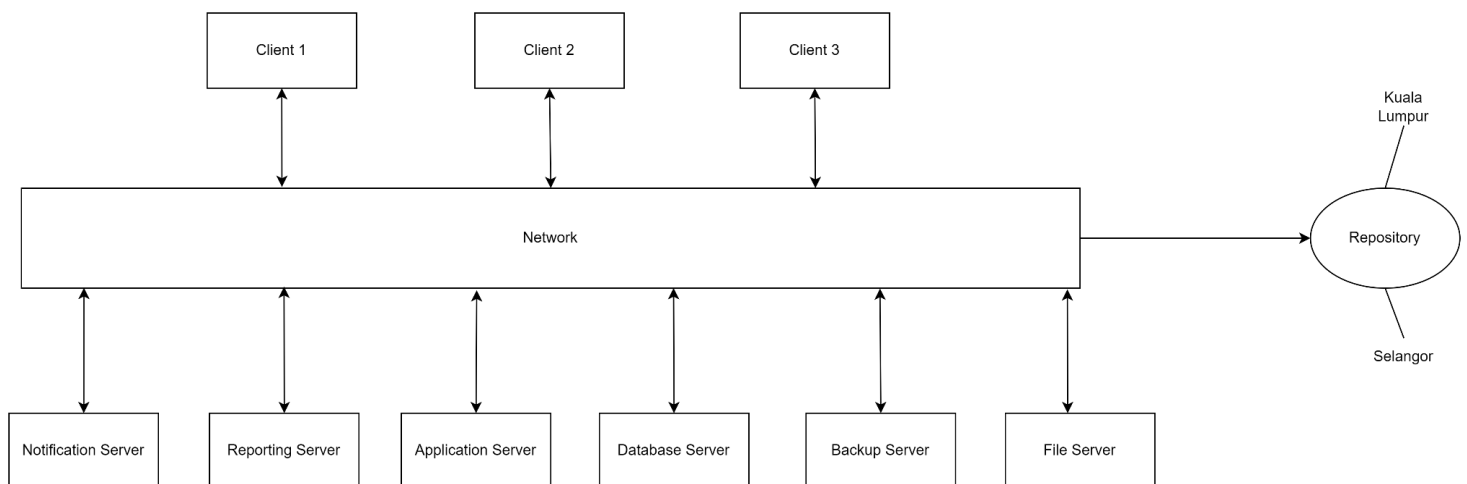
8.4 Parents (Gan Khai Li)

Parents play a crucial role as indirect users in the tuition centre's student information system, with responsibilities extending beyond mere oversight. Parents, as primary financial supporters of their children's education, ensure timely payment of tuition fees, which is essential for the student's continued enrollment and access to educational resources. They actively monitor their child's academic progress by keeping a close eye on grades, attendance records, and feedback from tutors. This enables them to stay informed about their child's performance and take necessary actions, such as scheduling or attending parent-teacher meetings to discuss their child's development. Additionally, parents are important participants in the system's feedback and communication processes. They provide valuable commentary on their child's experience at the tuition centre, offering input that helps improve both the learning process and the overall operation of the system. Their involvement underscores their role in guiding and assisting their child throughout their academic journey, ensuring that learning is both effective and successful.

8.5 Branch Managers (Cason Soh Kai Sheng)

The branch managers are solely responsible for the entire operation of the tuition centre from different branches. They can provide necessary features and functionalities needed for the system. The input of the branch managers' crucial understanding and perspective of the operation of the tuition centre will highly impact the system's overall efficiency and performance. The branch managers also serve as a liaison between all the stakeholders (teachers, students, parents), with that being said, the branch managers are able to provide various viewpoints and requirements. The branch managers are also mostly aware of the current trends in education and tutoring, their knowledge is also needed to influence the requirements needed in the system and to validate if the system can support long-term goals and objectives.

9.0 Architectural Design



This diagram above represents a network architecture where multiple clients interact with various servers, all of these components connect to a central repository. Here's a breakdown of the components and their roles:

1. Clients (Client 1, Client 2, Client 3):

- These are the end-user devices or systems that will be interacting with the network. They send and receive data through the network to perform various operations, such as requesting data, submitting information, or accessing services provided by the servers.

2. Network:

- This is the communication medium that connects all the clients and servers. It facilitates data transfer between the clients and the servers. The network ensures that all requests from the clients reach the appropriate servers and that the responses are sent back to the correct clients.

3. Servers:

- **Notification Server:** Manages notifications or alerts that may need to be sent to clients or other systems.
- **Reporting Server:** Handles generating and delivering reports based on the data processed or stored in the system.
- **Application Server:** Hosts and runs applications that the clients may need to interact with. It may handle the business logic and processing of client requests.
- **Database Server:** Manages the storage, retrieval, and querying of data. It interacts with the other servers and clients to provide data services.

- **Backup Server:** Handles the backup of data, ensuring that there are copies of important data that can be restored in case of failure or data loss.
- **File Server:** Manages file storage and sharing, allowing clients and other servers to store and access files.

4. **Repository:**

- This is a centralised storage system that holds data, files, or any other information needed by the clients and servers. In the diagram, it is shown to be connected to locations like "Kuala Lumpur" and "Selangor," which might indicate that this repository is accessed from or distributed across these locations.

5. **Kuala Lumpur & Selangor:**

- Represent geographical locations or specific branches/regions that interact with the repository. This suggests that the system is distributed across different locations, and data in the repository might be accessed or replicated across these areas.

Overall, this architecture depicts a distributed system where multiple clients interact with various servers, and all data or resources are centralised in a repository, which is accessible from different geographical locations.

10.0 Test Planning and Test Cases

10.1 Test Objective

The objective of testing the student and teacher subsystem is to ensure it meets all functional and non-functional requirements across different operating conditions. This testing will help to ensure that the subsystem integrates smoothly within the overall system, performs its intended functions accurately, and effectively manages edge cases and reduces potential failures.

10.2 Testing Process

10.2.1 Unit Testing

This is the first stage of the testing process where it involves testing each of the functions for all of our modules. The purpose of this is to verify that each function performs with expected result individually, ensuring that they produce the correct output after they are given specific inputs and that any internal logic is functioning correctly. For example, the register function where the personal information of a new user who registered recently will be recorded into the system correctly without any issue. Another example would be the search function where the search results are returned and displayed correctly.

10.2.2 Module Testing

Module testing is carried out right after unit testing. This stage involves testing the integration of functions within a module. The aim is to ensure that all functions interact correctly as individual modules. We have modules such as below.

Modules	Functions
User Management	Register, Search, Login, Delete, Modify
Course Management	Attendance Report, Timetable Management, Search, Real-Time Announcement Management, Calculate and Generate Report
Finance	Finance
Transport	Transport

All of the modules will be tested individually after integrating their respective functions. For example, the user management module will be tested by modifying the grade result of an account that was already registered by searching the account's username

10.2.3 Subsystem Testing

The subsystem testing is conducted after module testing. The purpose of this stage is to test the integration of multiple modules within a subsystem. The goal is to ensure that the subsystems function correctly with the expected outcome after combining their respective modules and interact seamlessly with each other. We have subsystems such as below.

Subsystem	Modules
User Management	User Management, Course Management
Course and Academic Management	Course Management
Financial Management Subsystem	Finance
Transport and Logistics Subsystem	Transport

To illustrate an example, the user management subsystem can be tested by logging into a recently registered account and using that account to update its attendance status for the day. The example involves the user management module and the course module.

10.2.4 System Testing

This stage of the testing process is the final stage of testing and it will be conducted right after the subsystem testing. This testing stage shifts the focus from testing individual subsystems to validating the integration and overall functionality of the entire system. The main goal of this testing stage is to ensure that all the subsystems are working together seamlessly as a whole while meeting the defined requirements. The final outcome of this testing stage would determine whether the software is for sure ready for deployment in the real environment or if there are any additional necessary adjustments that need to be made in order to resolve any issues that surface.

10.2.5 User Acceptance Testing

The user acceptance testing is conducted by having the actual stakeholders such as the staff and the students who will be chosen in order to ensure that the application meets all the necessary requirements and the users' expectations before it is released to the real world environment. It also helps when the software process model that the software will be referring to is the incremental model. The software will be developed and delivered in smaller, functional increments, where user acceptance testing will be performed on each increment as it is completed, unlike in a case where testing is performed right after the whole system is built. One of the main benefits of this is that the users can test the system early on and provide feedback in order to identify issues and make necessary changes and improvements. Feedbacks from earlier increments will then be incorporated into subsequent releases and in turn, enhancing overall system quality and user satisfaction. This would also help in reducing potential risks upon full system deployment. For example, the stakeholders can test the user management module by trying to perform any actions through the staff account such as modifying the grade of a student and make suggestions to improve the functions that they are testing on. The feedback will help the developers to make instant changes before moving on to the next module increment.

10.3 Tested Items

10.3.0 Functions Need To Be Tested

No.	Functions
1	Registration
2	Search
3	Login
4	Attendance Management
5	Calculate and Generate Report (Grade)
6	Modify and Remove Records
7	Timetable Management
8	Real-Time Announcement Management
9	Finance Management
10	Transport Booking

10.3.1 Function Which Not Be Tested

Maintenance Function

Functions such as system backup and server maintenance function will not be tested since they do not directly impact the main features or core system functionalities.

Optional User Features Function

Optional features such as customizable user preferences like theme colour and notification settings will be excluded from testing as they are not part of the core system functionality.

10.4 Testing Techniques

10.4.0 Unit Testing

10.4.0.0 White Box Testing

White box testing is a software testing technique that involves the testing of the internal logic and workings of the code. This requires the tester to have full access to the internal structure and the implementation of the code. This technique ensures that the code paths and conditions are executed as expected. This testing technique is well suited for unit testing due to the fact that it solely focuses on testing the internal structure and logic of the code. During unit testing, individual units such as functions, methods and classes will be tested in isolation. Each of their internal workings will be thoroughly tested. This would help to ensure that common mistakes such as logic errors and boundary conditions can be avoided early before the unit can be integrated with another unit to form a module. To illustrate an example, a function such as the calculation of students' final grades can be tested through white box testing by making sure its internal logic such as weighted score formula that was used and rounding standard are correct and producing expected result.

10.4.0.1 Bottom-up Testing

Bottom-up testing is a software testing technique where the testing starts from the lowest level components, which in this case is the unit testing and progresses to higher levels of integration. The units will be tested first before moving on to integrate them into modules. Bottom-up testing is one of the best approaches for unit testing due to the fact that the component unit is the most fundamental component. For example, the registration function can be tested first by creating a student account. Once the process and the outcome is validated, the login function will be the next in line to be tested by logging into the student account that was recently created after the registration process has been completed.

10.4.1 Module Testing

10.4.1.0 Black Box Testing

Black box testing is a software testing technique where the system's functionality will be evaluated by a tester without any knowledge of its code structure. The input values will be given to the tester to validate the output against expected results. This testing technique focuses on how a system behaves rather than how it is implemented. As modules are basically the combination of different functions, this proves that black box testing is best suited for module testing because the tester will focus on what the module should do and not how it does it. This ensures that the module delivers the right outputs when it is given a set of inputs while ignoring the internal logic of the module. For example, the course management module should allow the user, in this case the student, to take their attendance by entering the attendance code generated by the system according to their respective class following their schedules.

10.4.2 Subsystem

10.4.2.0 Thread Testing

Thread testing is another software testing technique where a bunch of functionally related modules are tested together under one subsystem in order to validate if they work as they should be when integrated. This testing technique focuses on verifying the flow of data and control through a specific transaction flow in the system which is to ensure that the subsystems interact correctly along that path. Thread testing is one of the best suited techniques for subsystem testing due to the fact that subsystems consist of multiple interconnected modules. Multiple modules within a subsystem can be validated by testing a thread which is a path through several modules. The fact that thread testing solely focuses on critical workflows, it is a big help in mitigating risk early by ensuring that key threads function properly. Risk of encountering major issues can be reduced significantly when more modules are integrating into the subsystem. For example, the user management subsystem can be tested by making sure that the input of a student's name in the search bar will also involve the attendance report module to generate a attendance status report of that specific student without any issues.

10.4.3 System

10.4.3.0 Stress Testing

Stress testing is a performance testing where the software system is put through under extreme workload conditions that are beyond its normal operating conditions. The reason for this testing is to determine its limit on how well it can handle in conditions such as high traffic, large volumes of data or with limited system resources. Stress testing is best suited for system testing because it helps to assess the system's overall stability under extreme circumstances. During system testing, the entire system is tested as a whole by stress testing it in order to verify that the integrated system can function as expected without any further issue surfaces even when the system is pushed beyond its limits. For example, a scenario where a lot of students' accounts are online at the same time accessing any of the system's features such as viewing timetables, updating attendance, or checking announcements that are available all at the same time. The number of students' accounts that will be tested altogether will keep increasing until the system reaches a breaking point. This scenario will test how well the system handles large numbers of simultaneous users accessing the system and its capabilities in overcoming it. Changes can be made based on the outcome of the test to further improve its capabilities and limits in order to overcome the scenario above.

10.4.3.1 Regression Testing

Regression testing is a type of software testing that ensures that recent changes or updates to a system such as new features added or new improvements made will not negatively affect the existing functionality of the system. Basically, regression testing is to perform testing right after fixes and updates have been made for the system, especially on the parts that could be affected by the changes made. Since system testing often follows major updates and fixes, regression testing is another well-suited testing technique in order to ensure that these changes will not unintentionally break any of the existing functionalities across the entire system. For example, after performing stress testing and applying the necessary updates and patches to the system, a regression testing can be performed to test all the main functionalities like updating student's grade, online payments and so on are not affected in any way. It is important to make sure that these critical functions are performed with intended outcomes and the system performance will not be negatively affected when tested under both normal and extreme conditions respectively.

10.5 Testing Environment

Testing Environment refers to the setup and configuration of resources such as hardware, software, network and so on that are necessary for executing test cases. It provides the controlled setting or environment in which the testing will be performed to ensure that the system performs with expected performance under conditions that imitate the real-world environment. To establish a comprehensive testing environment for mobile applications, specific hardware and software components are essential.

10.5.0 Hardware & Software

Every student, tutor and administrator use different mobile devices with different operating systems for their daily necessities. As such, it is crucial that the application is compatible with various types of devices and their specifications. The mobile devices that can be used as the testing environment are iPhone 13 with iOS 16, Samsung Galaxy S21 with Android 11 for the students. While on the other hand, devices with bigger screens such as iPad Pro with iOS 16 for the tutors and administrators. Devices like the iPhone 13 and Samsung Galaxy S21 represent a range of commonly used smartphones with different screen sizes and capabilities used by students. The iPad Pro provides a large screen and it is suitable for staff such as tutors and administrators to handle more complex tasks and data management.

Specification/Hardware	iPhone 13	Samsung Galaxy S21	iPad Pro
Screen Size (inches)	6.1	6.2	11
Resolution (pixels)	2532 x 1170	2400 x 1080	2388 x 1668
Processor	A15 Bionic Chip	Exynos 2100	M1 Chip
RAM (GB)	4	8	8
Storage (GB)	128	128	256
Battery Life (hours)	19	18	10
Network	5G	5G	5G
Operating System	iOS 16	Android 11	iOS 16

Screen size and resolution are important factors for testing how the application displays and caters to different screen sizes and resolutions, which would affect layout and usability. The choices of processor and RAM will help in evaluating the performance of the application under different hardware capabilities. This would allow us to ensure that the application can run smoothly on both

high-end and mid-range devices. The storage options affect how much data can be managed and stored locally. This is relevant for testing offline functionality if there is any in the future as well as data management purposes. The battery life indicates duration of use, which will impact performance and usability during prolonged testing sessions. Real-time features and data synchronisation can be tested with 5G network conditions.

10.5.1 Testing Tools

To automate the testing process, tools like Appium, which enables QAs to test apps automatically on several platforms, including IOS and Android. For platform-specific testing, Espresso can be used for Android applications, and XCUITest is suitable for IOS. Selecting performance testing tools is important as we can use these tools to assess the system's performance under load, stress and high traffic conditions. Example tools that can be used are Apache JMeter for load and stress testing and LordRunner for comprehensive performance testing. The developers can also use bug tracking tools such as JIRA for issue tracking and project management. These comprehensive tools ensure that all aspects of the mobile application, including performance, UI/UX, and compatibility across different devices and operating systems is thoroughly tested. By utilising these tools, a robust and reliable environment for software evaluation before deployment can be achieved.

10.6 Testing Constraints

1. Testing Schedule

The system should be tested during non-operational hours, specifically after 11:00 PM. This is to avoid any sort of occurrence of disruption on the daily activities of users of the applications such as students, teachers and administrators.

2. Compliance with Data Privacy Regulations

The testing must ensure that the system abides by data privacy regulations such as General Data Protection Regulation (GDPR). Data such as user data used in the tests must be replaced with dummy data to protect privacy.

3. Availability of Test Data

To ensure that the system can handle various real-world use cases effectively, a variety of realistic and diverse test data sets must be available to test the system across multiple scenarios. The data sets represent different student profiles, class schedules, examination results and attendance records.

4. Resource Limitations

Resources such as hardware, software and personnel should be considered during the testing environment process. Testing should be conducted and optimised for the resources available. Examples would be limited access to mobile devices and specific operating systems for compatibility testing.

5. Project Timelines and Release Schedules

Testing must adhere to project timelines and release schedules to ensure that all identified issues are resolved before deployment while also maintaining the release data commitments.

6. User Acceptance Testing Coordination

User Acceptance Testing should be coordinated with the availability of the students, tutors and administrators by informing them ahead of time to accommodate their schedules.

10.7 Test case

10.7.1 Search Function & Register Student Function

1. The staff is able to search the student information by entering the student's ID.
2. The staff is able to search the student information by entering the student's Name.
3. The staff is able to search the course information by entering the course Code.
4. The staff is not able to generate any search result if the input does not match any of the information in the system.
5. The student should be able to register a new account and be redirected to the login page.
6. The student should not be able to register a new account with the wrong format when entering their personal information

Tester's Name	Cason Soh Kai	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TCC mobile application
2	Sign up with a student account
3	Login with the student account
4	

S #	Test Data
1	Course Code = BACS2163
2	
3	
4	

Test Scenario	Test Search Function by Course Code
---------------	-------------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the search bar	The system shall allow user to enter an input in the search bar		
2	Enter Course Code	The system will allow the user to enter the course code		
3	Click the enter button or the search button	The system will display the course information that has the matched course code		

Tester's Name	Cason Soh Kai	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TCC mobile application
2	Sign up with a student account
3	Login with the student account
4	

S #	Test Data
1	Student Name = Cason Soh
2	
3	
4	

Test Scenario	Test Search Function by Student's Name
---------------	--

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the search bar	The system shall allow user to enter an input in the search bar		
2	Enter student name	The system will allow the user to enter the student name		
3	Click the enter button or the search button	The system will display the student information that has the matched student name		

Tester's Name	Cason Soh Kai Sheng	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to the TCC mobile application		1	Student ID = TTCSTU0001	
2	Sign up with a student account		2		
3	Login with the student account		3		
4			4		
Test Scenario					
Test Search Function by Student's ID					
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended	
1	Click on the search bar	The system shall allow user to enter an input in the search bar			
2	Enter student ID	The system will allow the user to enter the student ID			
3	Click the enter button or the search button	The system will display the student information that has the matched student ID			

Tester's Name	Cason Soh Kai Sheng	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to the TCC mobile application		1	Student ID = SSSSS	
2	Sign up with a student account		2	Student Name = Shu Han	
3	Login with the student account		3	Course Code = ABCS1234	
4			4		
Test Scenario					
Test Search Function if No Search Result Found					
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended	
1	Click on the search bar	The system shall allow user to enter an input in the search bar			
2	Enter student ID	The system will allow the user to enter the student ID			
3	Click the enter button or the search button	The system will display an error message "No search result found"			
4	Click on the search bar	The system shall allow user to enter an input in the search bar			
5	Enter student name	The system will allow the user to enter the student name			
6	Click the enter button or the search button	The system will display an error message "No search result found"			
7	Click on the search bar	The system shall allow user to enter an input in the search bar			
8	Enter course code	The system will allow the user to enter the course code			
9	Click the enter button or the search button	The system will display an error message "No search result found"			

Tester's Name		Cason Soh Kai Sheng	Date Tested				Test Case (Pass/Fail/Not				
S #	Prerequisites:			S #	Test Data						
1	Access to the TCC mobile application			1	Name = Cason Soh						
2	In the Setup Page, click "Get Started" button			2	Branch = Kuala Lumpur						
3	In the Login Page, click "Register"			3	D.O.B = 29.02.2004						
4				4	IC No. = 123456-69-6666						
				5	Phone No = +60 11-12345678						
				6	Address = 69, Triangle, Jalan Indah, 52000, Kuala Lumpur						
				7	Email = cason29@gmail.com						
				8	Nationality = Cina						
				9	Password = 123456789						
				10	Reenter Password = 123456789						
Test Scenario		Registration Success									
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended			
1	Click the Name column		The system will allow the user to enter input in the Name column								
2	Enter name		The system will allow the user to enter name								
3	Click the Branch column		The system will allow the user to view a dropdown list containing branch names								
4	Select Branch		The system will allow the user to select branch								
5	Click D.O.B column		The system will allow the user to enter input in the D.O.B column								
6	Enter D.O.B		The system will allow the user to enter D.O.B								
7	Click IC No. column		The system will allow the user to enter input in the IC No. column								
8	Enter IC No.		The system will allow the user to enter IC No.								

9	Click Phone No. column	The system will allow the user to enter input in the Phone No. column		
10	Enter Phone No.	The system will allow the user to enter Phone No.		
11	Click Address column	The system will allow the user to enter input in the Address column		
12	Enter Address	The system will allow the user to enter Address		
13	Click Email column	The system will allow the user to enter input in the Email column		
14	Enter Email address	The system will allow the user to enter Email address		
15	Click the "Next" button	The system will redirect the user to the next page		
16	The User ID will be autogenerated	The system will autogenerate a User ID once "Next" button has been clicked		
17	Click Nationality column	The system will allow the user to view a dropdown list containing nationality names		
18	Select Nationality	The system will allow the user to select nationality		
19	Click Password column	The system will allow the user to enter input in the Password column		
20	Enter Password	The system will allow the user to enter Email address		
21	Click Reenter Password column	The system will allow the user to enter input in the Reenter Password column		
22	Reenter password	The system will allow the user to enter password		
23	Click on the "Register" button	The system will redirect the user to the Login Page		

Tester's Name		Cason Soh Kai Sheng	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data		
1	Access to the TCC mobile application		1	D.O.B = 29 FEB 2004		
2	In the Setup Page, click "Get Started" button		2	IC No. = 123456696666		
3	In the Login Page, click "Register"		3	Phone No = 01112345678		
4			4	Email = cason29@gmailcom		
			5	Password = 123456789		
			6	Reenter Password = 987654321		
Test Scenario		Registration Unsuccess				
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
5	Click D.O.B column	The system will allow the user to enter input in the D.O.B column				
6	Enter D.O.B	The systme will allow the user to enter D.O.B				
	Click any column	The systme will display an error message "Wrong format, please type again."				
7	Click IC No. column	The system will allow the user to enter input in the IC No. column				
8	Enter IC No.	The systme will allow the user to enter IC No				
	Click any column	The systme will display an error message "Wrong format, please type again."				
9	Click Phone No. column	The system will allow the user to enter input in the Phone No. column				
10	Enter Phone No.	The systme will allow the user to enter Phone No.				
	Click any column	The systme will display an error message "Wrong format, please type again."				
15	Click the "Next" button	The system will redirect the user to the next page				

16	The User ID will be autogenerated	The system will autogenerate a User ID once "Next" button has been clicked		
21	Click Password column	The system will allow the user to enter input in the Password column		
22	Enter Password	The systme will allow the user to enter Email address		
23	Click Reenter Password column	The system will allow the user to enter input in the Reenter Password column		
24	Reenter password	The system will allow the user to enter password		
25	Click on the "Register" button	The system will display an error message "Password is not the same, please try again."		

10.7.2 Calculate and Generate Reports for Grades Function & Remove/Edit Existing Record Function

1. The staff should not be able to modify the information with the wrong format
2. Search the report, successfully generate the report that selected
3. Search the report that is don't have the report in that year or month, it will generate the error message
4. When input the negative value to calculate the report of grades, it will not calculate it due to the negative number
[Quiz1, Quiz2, Test, Mock Test]

Tester's Name	Hea Hui Min	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TTC mobile application
2	Login with admin account
3	
4	

S #	Test Data
1	IC : 091408-15-0055
2	IC : 091408-15-00555
3	
4	

Test Scenario	Modify student IC
---------------	-------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click Student Information	The system will display all the student		
2	Click the 'Modify' button	The system display all the information of student and can able to choose and modify it		
3	Choose the student information that need to edit	The system shall allow the user to enter the value		
4	Click the "Submit" button	The system will display the error message that this data that new modify s not follow the format		

Tester's Name	Hea Hui Min	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to the TTC mobile application		1	Year : 2022	
2	Login with student account		2	Month : MAY	
3			3		
4			4		

Test Scenario	Input the Year of report that is 2022 MAY
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click the search bar	The system display a space to let user to enter the subject name		
2	Click the "Year" button	The system shall allow user to select the year		
3	Select the year	The system shall display all the year selected exam result		
4	Click the "Month" button	The system shall allow user to click the month		
5	Select the month	The system shall display all the month selected exam result		

Tester's Name		Hea Hui Min	Date Tested				Test Case (Pass/Fail/Not Executed)		
S #	Prerequisites:					S #	Test Data		
1	Access to the TTC mobile application					1	Year : 2019		
2	Login with student account					2	Month : MAY		
3						3			
4						4			
Test Scenario		Select the Year of result that is not such a result found							
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended		
1	Click the search bar		The system display a space to let user to enter the subject name						
2	Click the "Year" button		The system shall allow user to select the year						
3	Select the year		The system will prompt out the error message "No Such A Result Found"						
4	Click the "Month" button		The system shall allow user to click the month						
5	Select the month		The system will prompt out the error message "No Such A Result Found"						

Tester's Name		Hea Hui Min	Date Tested			Test Case (Pass/Fail/Not Executed)		
S #		Prerequisites:			S #		Test Data	
1		Access to the TTC mobile application			1		Student Name : Michalle	
2		Login with teacher account			2		Subject : Science	
3					3		Quiz1: 87 Quiz2: -90 Test : 78 Mock Test : 80	
4					4			
Test Scenario		Input the negative value						

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click the 'input marks' button	The system display the table		
2	Click the Student Name	The system shall allow user to enter the student name		
3	Click the subject	The system shall allow user to choose the subject		
4	Input the test data	The system display the error message 'Mark don't have negative value'		

10.7.3 Login Function & Attendance Management Function

1. The student should be able to login to their account with valid ID and password
2. The student should not be able to login to the account if they entered the invalid ID or password.
3. The student should be able to reset the password if their reset password is match with the password confirmation after entering their User ID and Identification Card Number
4. The student should not be able to reset the password if their reset password is mismatched with the password confirmation that is entered or the pattern of password does not meet the requirements that are needed.
5. The student should be able to take their attendance with the correct attendance PIN code.
6. The student should not be able to take their attendance using the wrong attendance PIN code
7. The student should not be able to take the same class attendance twice

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TCC mobile application
2	Register with a student account
3	
4	

S #	Test Data
1	UserID = TTCSTU0001 Password = mich@12345
2	UserID = TTCSTU0001 Password = mich@12345
3	
4	

Test Scenario	Test Login Credential
---------------	-----------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click the User ID column	The system should allow the user to enter the user ID		
2	Enter User ID	The system should allow the user to enter the password		
3	Enter the password	The system should allow the user to click on the login button		
4	Click on the login button	The user should login successfully and the login successfully message is prompt		

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not	
---------------	-----------------	-------------	--	--------------------------	--

S #	Prerequisites:
1	Access to the TCC mobile application
2	Register with a student account
3	
4	

S #	Test Data
1	UserID = TTCSTU0001 Password = mich@12345
2	UserID = TTCSTU0001 Password = mich@12111
3	UserID = TTCSTU Password = mich@12345
4	

Test Scenario	Test Invalid Login Credential
---------------	-------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click the User ID column	The system should allow the user to enter the user ID		
2	Enter User ID	The system should allow the user to enter the password		
3	Enter the invalid password	The system should allow the user to click on the login button		
4	Click on the login button	The system should display invalid login and request the user to login again		

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not	
---------------	-----------------	-------------	--	--------------------------	--

S #	Prerequisites:
1	Access to the TCC mobile application
2	Register with a student account
3	
4	

S #	Test Data
1	User ID = TTCSTU0001 IC Number = 080808-14-1521 Password reset = mich@12333 Password confirmation = mich@12333
2	
3	
4	

Test Scenario	Test on reset the password
---------------	----------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the forgot password icon	The system should allows user to enter their User ID and Identification Card Number to change password and click on confirm		
2	Enter their User ID and Identification Card Number and click on confirm	The system should provide a field to allows users to enter their new the password that is at least 8 characters, including an numeric character and special character		
3	Enter a new password	The system will display a message to request users to reenter the password		
4	Reenter the password	The password is reset successfully and a message will prompt to tell user successfully in reset password		
5				

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TOC mobile application
2	Register with a student account
3	
4	

S #	Test Data
1	User ID = TTCSTU0001 IC Number = 080808-14-1521 Password reset = mich@12333 Password confirmation = mich@12333
2	User ID = TTCSTU0001 IC Number = 080808-14-1521 Password reset = mich@12333 Password confirmation = mich@123334
3	User ID = TTCSTU0001 IC Number = 080808-14-1521 Password reset = mich@12333 Password confirmation = 123334
4	

Test Scenario	Test on invalid reset the password
---------------	------------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the forgot password icon	The system should allows user to enter their User ID and Identification Card Number to change password and click on confirm		
2	Enter their User ID and Identification Card Number and click on confirm	The system should provide a field to allows users to enter their new the password that is at least 8 characters, including an numeric character and special character		
3	Enter a new password	The system will display a message to request users to reenter the password		
4	Reenter the password for confirmation	The password mismatch with the previous reset password or does not meet the requirement of the password pattern. The system will prompt invalid password and allows users to reenter the reset password		

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TOC mobile application
2	Register with a student account
3	Login with student valid user ID and password
4	

S #	Test Data
1	Attendance Code = 347680
2	
3	
4	

Test Scenario	Test Enter Valid Attendance Code
---------------	----------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the Attendance icon	The system should display the attendance code page and allows users to enter the attendance pin code		
2	Enter the attendance pin code	The system display the success and confirm message to the user		
3	Click on the attendance report icon	The attendance should be recorded and display		
4				

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:
1	Access to the TCC mobile application
2	Register with a student account
3	Login with student valid user ID and password
4	

S #	Test Data
1	Attendance Code = 347680
2	Attendance Code = 789567
3	
4	

Test Scenario	Test Enter Invalid Attendance Code
---------------	------------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the Attendance icon	The system should display the attendance code page and allows users to enter the attendance pin code		
2	Enter the attendance pin code	The system display invalid PIN code		
3	Click on the attendance report icon	The attendance PIN code should not be displayed		
4				

Tester's Name	Ethel Ng Yi Yan	Date Tested		Test Case (Pass/Fail/Not Executed)	
---------------	-----------------	-------------	--	------------------------------------	--

S #	Prerequisites:
1	Access to the TCC mobile application
2	Register with a student account
3	Login with student valid user ID and password
4	Attendance Code has been taken

S #	Test Data
1	Attendance Code = 347680
2	Attendance Code = 347680
3	
4	

Test Scenario	Test Enter Duplicate Attendance Code
---------------	--------------------------------------

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Click on the Attendance icon	The system should display the attendance code page and allows users to enter the attendance pin code		
2	Enter the attendance pin code	The system display message to user that the attendance code has been taken		
3	Click on the attendance report icon	The system will prompt a message to tell that the record has exists		
4				

10.7.4 Timetable Management Function & Real Time Announcement Management Function

1. The system identifies a conflict when a teacher is assigned to overlapping classes.
2. The student can successfully log in and view their personalised class schedule.
3. The admin can successfully edit and update an existing schedule to accommodate a replacement class.
4. After posting an announcement, the website successfully displays content that matches the posted information.
5. The system tracks and records whether users have read the announcement.
6. The admin can delete an existing announcement from the system.

Tester's Name		Gan Khai Li	Date Tested				Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:				S #	Test Data				
1	Access to the TTC mobile application				1	AdminID: TTCADM0001 AdminName: John Password: John@54321				
2	Login with the admin account				2	TeacherID: TTCTE0001 TeacherName: Mr. Davis Subject: Math				
3	Teacher's schedules already exist				3	Existing Schedule: Thursday Time: 2:00PM-4:00PM Venue: Room101				
4					4	New Class Details: Thursday Time: 2:30PM-4:30PM Venue: Room202				
Test Scenario		Verify Conflict Detection for Overlapping Teacher Schedule								
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended			
1	Log in as an admin		The admin should be successfully logged in.							
2	Navigate to the "Create Schedule" section		The "Create Schedule" section should be displayed.							
3	Attempt to schedule a new class for a teacher at a time when they are already assigned to another class		The system should display an error message indicating a scheduling conflict for the teacher.							
4	Save the schedule		The system should prevent saving the schedule and notify the user about the clash.							

Tester's Name		Gan Khai Li	Date Tested				Test Case (Pass/Fail/Not Executed)			
S #		Prerequisites:		S #		Test Data				
1		Access to the TTC mobile application		1		StudentID:TTCSTU0001 Password:mich@12345				
2		Login with the student account		2		StudentID:TTCSTU0001 StudentName: Michalle Class List: Account: Monday (4:00PM-6:00PM, Room 202) English: Tuesday (3:30PM-5:30PM,Room 102)				
3		Student's schedules already exist		3						
4				4						
Test Scenario		Verify Student's Ability to View Personalized Class Schedule								
Step #		Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended		
1		Log in to the system as a student		The student should be successfully logged in and redirected to their dashboard.						
2		Navigate to the "View Schedule" section		The "View Schedule" section should be displayed, showing the student's timetable for the current day or week.						
3		Review the displayed class schedule.		The schedule should correctly list all classes assigned to the student, including class names, times, and venue. It should match the provided test data.						
4		Compare the displayed timetable details with the expected schedule		Timetable details match the records in the system.						

Tester's Name		Gan Khai Li	Date Tested				Test Case (Pass/Fail/Not Executed)				
S #		Prerequisites:		S #		Test Data					
1		Access to the TTC mobile application		1		AdminID: TTCADM0001 AdminName: John Password: John@54321					
2		Login with a admin account		2		Existing Schedule: ClassName:English Date: 10-Oct-2025 Time:3:30 PM-5:30 PM Venue: Room102					
3		The schedule that needs to be edited already exists		3		Replacement Class Details: Date: 12-Oct-2025 Time:2:00 PM-4:00 PM Venue: Room103					
4		Replacement schedule already assigned		4							
Test Scenario		Verify Admin's Ability to Edit and Update Schedule for Replacement Class									
Step #		Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended			
1		Log in to the system as an admin.		The admin should be successfully logged in and redirected to the dashboard.							
2		Navigate to the "Edit Schedule" section.		The "Edit Schedule" section should be displayed with a list of existing schedules.							
3		Select an existing schedule that needs to be modified for a replacement class.		The existing schedule details should be displayed, allowing edits.							
4		Modify the schedule details, such as changing the time, date, or room for the replacement class.		The system should allow the admin to make the necessary changes.							
5		Click the "Update" button to save the changes.		The system should update the schedule, and the changes should be reflected on the timetable.							
6		Verify that the changes are accurately reflected in the timetable.		The updated schedule should show the replacement class details correctly .							

Tester's Name		Gan Khai Li	Date Tested				Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:				S #	Test Data				
1	Access to the TTC mobile application				1	AdminID: TTCADM0001 AdminName: John Password: John@54321				
2	Login with a admin account				2	Announcement Title: "Holiday Notice" Announcement Content: "The center will be closed on 16th September for National Day."				
3	In the Announcement Page, click "Create"				3	Audience: "All Users"				
4					4	Posting Date: 10-Sep-2024 Posting Time: 09:00 AM				
Test Scenario		Verify Announcement Appears on Website After Posting								
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended		
1	Log in to the system as an admin.		The admin should be successfully logged in and redirected to the dashboard.							
2	Navigate to the "Create Announcement" page via the admin panel.		The "Create Announcement" page should open successfully, allowing the admin to enter details for the new announcement.							
3	Enter the announcement title, content, select the audience, and set the posting date and time.		The system should accept the input data and validate it without errors.							
4	Submit the announcement by clicking the "Post" or "Save" button.		The system should display a confirmation message indicating that the announcement has been successfully posted.							
5	Open the website and navigate to the announcements section.		The newly posted announcement should be visible on the website, under the announcements section, immediately or at the scheduled time.							
6	Verify the announcement's title, content, and audience match the details entered by the admin.		The announcement displayed on the website should match exactly with the title, content, and audience that was specified during creation.							

Tester's Name	Gan Khai Li	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to the TTC mobile application		1	Regular User Account: Username: testuser Password: password123	
2	The announcement tracking feature has been enabled		2	AdminID: TTCADM0001 AdminName: John Password: John@54321	
3	Login with Valid User ID and Password		3	Announcement ID: ANN001 Title: "System Maintenance Scheduled" Content: "System maintenance is scheduled for this Sunday 12:00AM-8:00AM. Please save your work."	
4	At least one announcement has existed in the system		4	Status Tracking: Initial status is Unread for all users.	
Test Scenario Verify System Tracks and Records User Read Status for Announcements					

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Log in as a regular user and navigate to the announcements section.	Announcements should be visible, and their status should be tracked. Verify that the announcement is listed and shows the initial status as "Unread".		
2	Click on an announcement to mark it as read.	The system should record the user's action and mark the announcement as read. The system should reflect this change immediately or within an acceptable time frame.		
3	Log in as an admin and check the reading status report for the announcement.	The admin should see the status (read/unread) for each user regarding the specific announcement.		
4	Verify if the read/unread status is accurately updated and reflected in the admin report.	The status in the report should match the actions performed by the user. Make sure the status matches the user's actions and has no mistakes.		

Tester's Name	Gan Khai Li	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to the TTC mobile application		1	AdminID: TTCADM0001 AdminName: John Password: John@54321	
2	Login with the admin account		2	Title: "Holiday Update"	
3	Announcements already exist		3		
4	In the Announcement Page, click "Delete"		4		
Test Scenario Verify Delete Announcement for Admin					

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Log in to the system as an admin.	The admin should be successfully logged in and redirected to the dashboard.		
2	Navigate to the "Announcements" section.	The page should open as expected.		
3	Select the announcement titled "Holiday Update"	The correct announcement is selected.		
4	Click the "Delete Announcement" button for the selected announcement	A confirmation dialog should appear.		
5	Confirm deletion by clicking "Confirm" on the dialog	The announcement should be deleted and no longer visible in the list.		

10.7.5 Transport Booking Function & Payment Management Function

1. The student cannot be assigned to two vehicles at the same time
2. The student cannot cancel their transportation booking after 7PM on the day before the scheduled service.
3. Students will receive notification after successful payment
4. The student can successfully book transportation for a round trip from home to tuition centre and back home.
5. After payment successful, generate invoice to the student and stored in the invoice history and update the status of payment
6. The system prevents the same transaction from being processed multiple times, ensuring no duplicate payments.

Tester's Name	Kit Chin Jie Ying	Date Tested		Test Case (Pass/Fail/Not Executed)	
S #	Prerequisites:		S #	Test Data	
1	Access to Chrome/Safari Browser		1	Student ID: TTCSTU0001 Existing Vehicle: VAD 1234 Existing Service Time: 8AM - 9AM New Vehicle: VAD 5678 New Service Time: 8AM - 9AM	
2	Access to the transportation management system		2	Student ID: TTCSTU0002 Existing Vehicle: VAD 5678 Existing Service Time: 8AM - 9AM New Vehicle: VAD 1234 New Service Time: 8AM - 9AM	
3	A student is already assigned to a vehicle for a specific time slot		3		
4			4		
Test Scenario	Verify that a student cannot be assigned to two vehicles at the same time				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended	
1	Navigate to the scheduled transportation services page	The page should be open as expected			
2	Attempt to assign the student to a new vehicle with the same service time as an existing assignment	The system should detect the conflict and prevent assignment			
3	Verify that an error message is displayed indicating a schedule conflict	An appropriate error message should be shown ("Student is already assigned to another vehicle at this time")			
4	Attempt to assign the student to a different vehicle at a non-overlapping time	The assignment should be successful without any conflict			

Tester's Name		Kit Chin Jie Ying	Date Tested				Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data					
1	Access to Chrome/Safari Browser			1	Student ID: TTCSTU0001 Transportation Request ID: TR001 Current Date and Time: DATE, 7.30PM					
2	Access to the transportation booking system with user credentials			2	Student ID: TTCSTU0002 Transportation Request ID: TR002 Current Date and Time: DATE, 7.00PM					
3	A scheduled transportation service for the next day			3						
4	The current time is after 7 PM on the day before the scheduled service			4						
Test Scenario										
Verify that the system prevents the student from canceling their transportation after 7 PM on the day before the scheduled service.										
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended			
1	Navigate to the scheduled transportation booking page		The page should be open as expected							
2	Attempt to cancel the scheduled transportation service after 7 PM		The system should detect the conflict and prevent assignment							
3	Verify that an error message is displayed		An appropriate error message should be shown ("Cancellation is not allowed after 7 PM the day before the service")							
4	Attempt to cancel the transportation service before 7 PM		The system should allow the cancellation successfully							
5	Verify that the cancellation is processed successfully before 7 PM		The transportation service should be canceled without issues.							

Tester's Name		Kit Chin Jie Ying	Date Tested				Test Case (Pass/Fail/Not Executed)				
S #		Prerequisites:			S #		Test Data				
1		Access to Chrome/Safari Browser			1		Student ID: TTCSTU0003 Valid Payment Method: Credit Card ending in 6666 Payment Amount: RM321				
2		Access to the payment system with user credentials			2		Student ID: TTCSTU0004 Valid Payment Method: Touch N Go Payment Amount: Rm442				
3		A valid payment method linked to the user's account			3						
4		The system is configured to send notifications for payment events			4						
Test Scenario		Validate that users receive a notification after successful payments									
Step #		Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended			
1		Navigate to the payment page and select a service to pay for		The page should be open and allow selection of the service							
2		Complete the payment using the valid payment method		The payment should be processed successfully							
3		Verify that a success notification is displayed on-screen		A confirmation message should appear, indicating successful payment.							
4		Check for any notification or email confirmation sent to the user		The user should receive a notification or email confirming the successful payment ("Thank you for your payment.").							

Tester's Name		Kit Chin Jie Ying	Date Tested				Test Case (Pass/Fail/Not Executed)				
S #		Prerequisites:				S #		Test Data			
1		Access to Chrome/Safari Browser				1		Student ID: TTCSTU0005 Date of Transportation: 12-10-2024 Pick up Time: 8am Return Time: 12pm Home Address: 123, Jalan Maple, Taman Maple, 43200, Cheras, Selangor.			
2		Access to the transportation booking system with user credentials				2		Student ID: TTCSTU0006 Date of Transportation: 16-10-2024 Pick up Time: 10am Return Time: 11am Home Address: A-09, Jalan Satu, Taman Bunga, 43200, Cheras, Selangor.			
3		Available transportation options for both the trip to the center and the return trip.				3					
4						4					
Test Scenario		Verify that a student can successfully book transportation for a round trip from home to the tuition center and back home.									

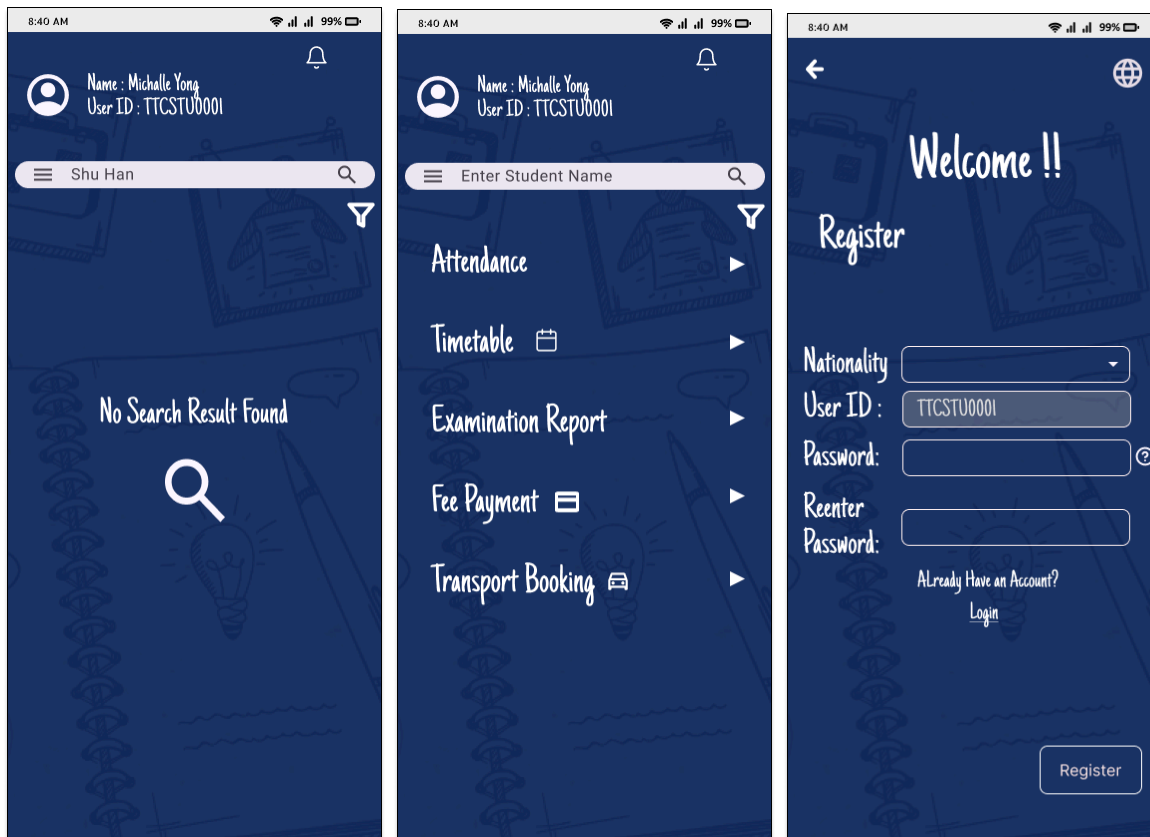
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended	
1	Navigate to the transportation booking page		The page should open and display available booking options					
2	Choose the desired pickup time (e.g: 8AM) and confirm the one-way trip booking		The system should confirm the one-way booking and display the trip details.					
3	Choose the desired return time (eg: 12 PM) and confirm the round-trip booking		The system should confirm the booking for both trips and show the combined trip details for both the outbound and return trips					
4	Verify that the round trip details are correctly listed in the student's booking history		The booking history should list the round trip with correct timings and details for both legs of the trip.					
5	Verify that a confirmation notification is sent to the student		The student should receive a notification confirming the round trip booking					

Tester's Name		Kit Chin Jie Ying	Date Tested				Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:				S #	Test Data				
1	Access to Chrome/Safari Browser				1	Student ID: TTCSTU0006 Date of Payment: 5-9-2024 Payment amount: RM200				
2	Access to the payment system with user credentials				2	Student ID: TTCSTU0007 Date of Payment: 1-9-2024 Payment amount: RM132				
3	A pending fee or service charge that needs to be paid.				3					
4	The ability to generate, view, and store invoices within the system.				4					
Test Scenario		Verify that after the user pays the fee, an invoice is generated and stored in the invoice history, and the payment status is updated to "Paid."								
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended			
1	Navigate to the payment system		The payment system should open, displaying any pending fees							
2	Select the pending fee to be paid		The system should allow the selection of the pending fee and proceed to payment options							
3	Complete the payment using the selected payment method		The payment should be processed successfully, and a confirmation message should appear indicating the payment is complete.							
4	Verify that an invoice is generated for the payment		The system should generate an invoice with the details of the transaction							
5	Check the invoice history to ensure the newly generated invoice is stored		The invoice should be listed in the invoice history with correct details							
6	Verify that the payment status for the transaction is updated to "Paid"		The payment status should be updated to "Paid" in the system, reflecting the successful completion of the transaction.							
7	Confirm that the payment history reflects the updated status		The payment history should show the transaction as completed, with the correct payment status and linked to the invoice							

Tester's Name		Kit Chin Jie Ying	Date Tested			Test Case (Pass/Fail/Not Executed)		
S #	Prerequisites:			S #	Test Data			
1	Access to Chrome/Safari Browser			1	Student ID: TTCSTU0007			
2	Access to the payment system with user credentials			2	Initial Payment Status: "Paid"			
3	A transaction that has already been processed and marked as "Paid."			3				
4	System functionality to prevent duplicate payments.			4				
Test Scenario		Verify that the system prevents the same transaction from being processed multiple times, ensuring no duplicate payments.						
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended	
1	Navigate to the payment system		The payment system should open, displaying completed transactions					
2	Navigate to the payment history page		The page should show a list of all past transactions, including those marked as "Paid."					
3	Select a transaction that has already been marked as "Paid"		The system should show that the transaction has already been completed					
4	Attempt to initiate the payment again for the same transaction		The system should prevent the duplicate payment and display an error message such as "This transaction has already been paid."					
5	Verify that no duplicate payments are recorded in the payment history		The payment history should remain unchanged, showing the transaction as "Paid" without additional entries for duplicates.					

11.0 User Interface Design Principles

11.1 Search and Registration Functions (Cason)



?

Password Settings

X

- The password includes at least 1 numeric character, 1 alphabet and 1 special character
- The password must be at least 8 characters in length

- **User Diversity**

- Different users have different needs, preferences and levels of expertise. The interface should be flexible in order to cater to a wide variety of users, from beginners to experts. This would help to ensure accessibility for all. The registration page is splitted into two pages and it introduces a clear. Step-by-step registration process. This segmentation would accommodate users who may need more time to fill in forms. This is to make sure that the users are not overwhelmed by too many fields in a single page. The simplicity of the forms would help to cater to both tech-savvy experienced users and users who have less experience.

- **User Familiarity**

- The user interface should use concepts, terminology and visual elements that the users are familiar with. This would leverage the users previous knowledge and experiences, which could help reduce the learning curve while at the same time making the interface more intuitive. The registration pages use standard form fields for input such as “Name”, “Date of Birth”, “Phone No.”, and “Email” which are universally recognized fields in most of the registration forms. The usage of common icons for actions such as back navigation (top left arrow) and toggling options (drop-down arrow) which builds on users’ pre-existing knowledge from other apps that they have used before, which makes the navigation intuitive.

- **User Guidance**

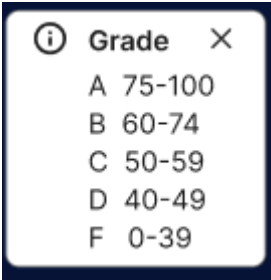
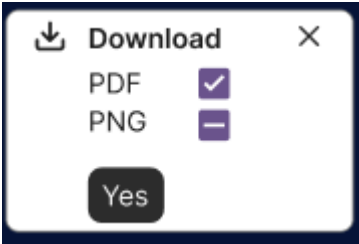
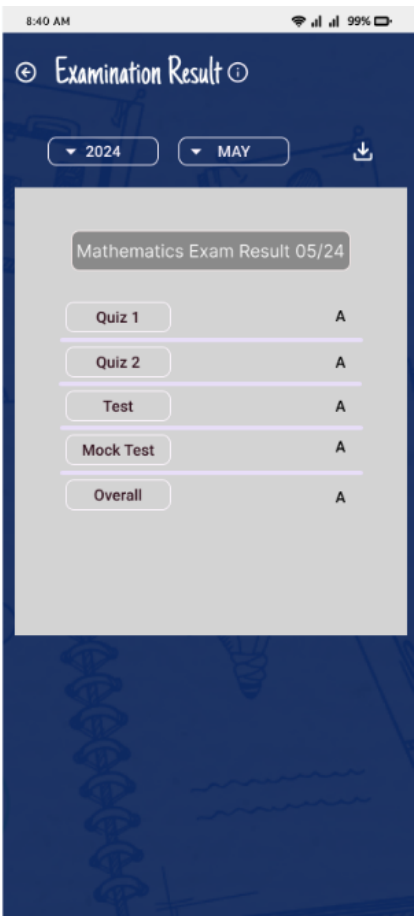
- The user interface should provide clear instructions, feedback, and assistance to users while they are navigating through the system. All of this can be achieved by using tooltips, prompts and visual cues to ensure that the users are aware of their next steps and will have no trouble in understanding the system’s behaviour. To put up examples, the user interface design provides guidance by using labels for each input field to ensure that users will know exactly what information they have to provide. The usage of buttons with labels such as “Next” and “Register” clearly indicate the action that will follow. In addition, the “Already have an Account?” link guides users who may be trying to just login instead of registering, directly to the login page.

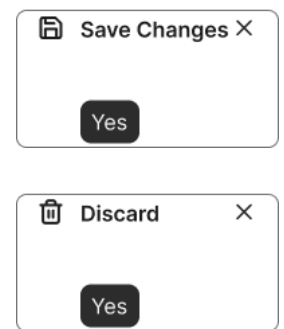
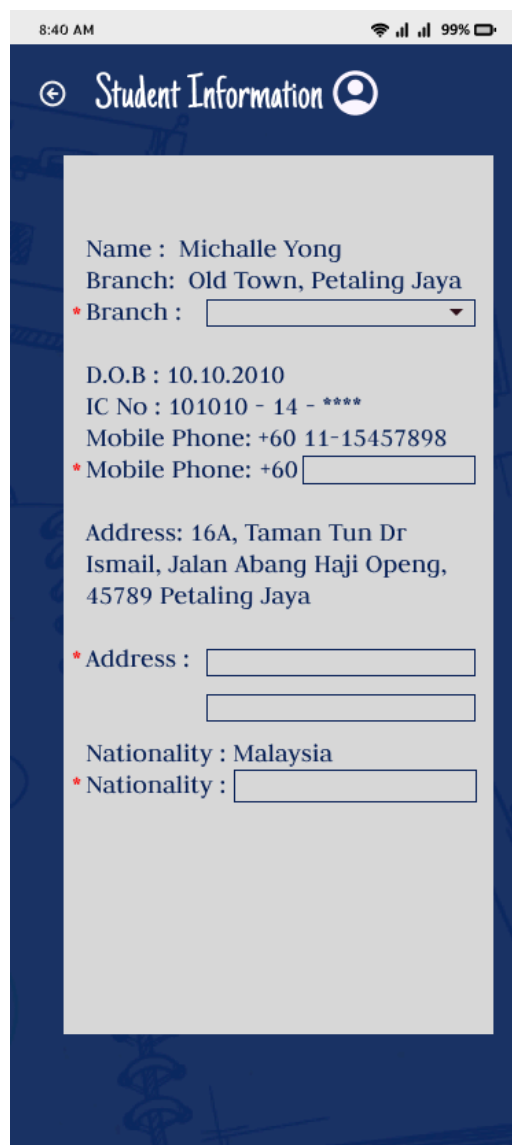
- **Minimal Surprises**

- The system’s behaviour should be predictable without having any unexpected outcomes from occurring. The users will feel in control and confident while interacting with the interface when the design and functionality of the user interface is consistent. The user interface designs’ layout and actions are consistent. The transition from the first registration page to the second page keeps the same structure

which will minimise surprises for the user. The main page follows a logical, menu-driven structure for accessing key functionalities such as “Attendance”, “Timetable”, and “Examination Report.” This would ensure that the users will not encounter any sort of unexpected elements that could disrupt their user experience.

11.2 Remove/Edit Records and Generate Report(Grade) Functions (Hea Hui Min)





- **User Familiarity**

- The application has the same familiar icon such as save icon and the discard button. The arrow left icon means that the user needs to go back to the previous page, the downward arrow icon indicates the user can drop down a list to view and select, the dustbin icon indicates the user needs to abandon changes, while the file icon indicates that the user wants to save changes.

- **User Guidance**

- The application design also provides the user guidance such as the icon introduction. When we click on the introduction icon in the test results, detailed information about the examination results will be displayed. For example, the score range corresponding to A level is 75-100 points. Furthermore, to ensure used fill in the information correctly, our app using red * marks to indicate the fields are required to provide clear guidance when modifying student information.

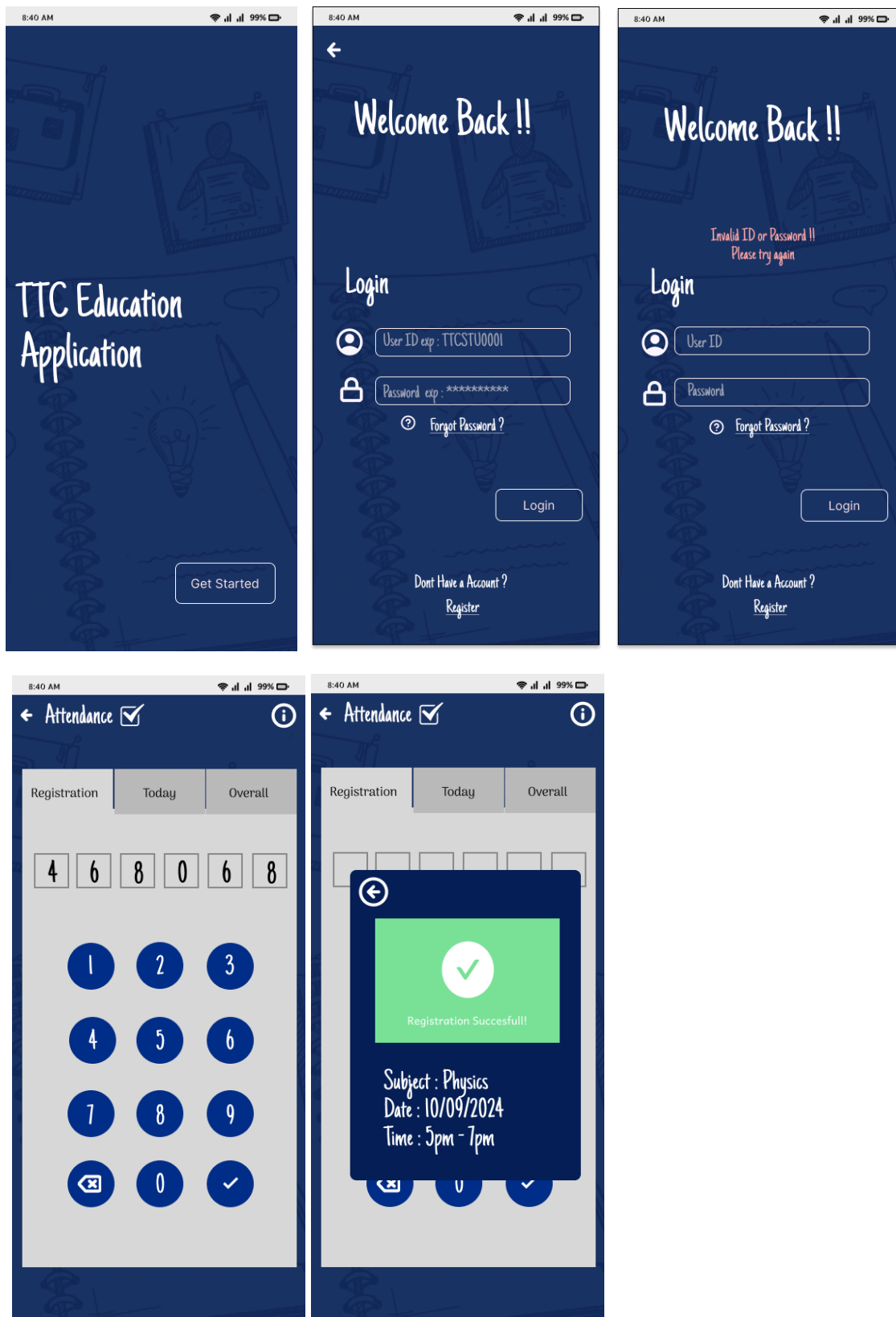
- **Minimal Surprise**

- The application design also is the minimal surprise which is the user's expectation .When I click the save button it will prompt out the message to inform the user whether to save the changes or not. If not, click the cancel button to go back to the edit page. When the user decides there is no need to edit already, it can click the discord button to go back to the student profile information page.

- **Consistency**

- The application design also provides consistency such as all the formats of mobile phones start from +06. This is because the Malaysia phone number is default is +06. Besides, all the font style, size inside the student information are the same to let the information not be messy. It is important to note that the 'Branch', 'Year' and 'Month' options are fixed and users can only choose from the present options.

11.3 Login and Attendance Function (Ethel)



 **Hint** 

- The password includes at least 1 numeric character, 1 alphabet and 1 special character
- The password must be at least 8 characters in length

 **Attendance Guide** 

- You can only check your overall attendance for the current month only
- You may only register your attendance during the class period.
- Attendance Code will be given by the tutor during class period

- **User Familiarity**

- This application has the same iconic icon that is familiar to the user so the user will not be confused while navigating the application. For example, the back button on the top left to represent going back or moving back to the previous page. Besides that we also used the same layout for login credentials as other famous website user id on top of the password field, which allows users to easily learn without extra learning time.

- **Recoverability**

- The application that was designed will not punish users when it makes some silly or accidental mistake. By either allowing them to take back an action. This can be seen when the application allows the user to use the forget password function to reset the user credentials. Besides that, for the attendance function, the system allows users to undo the attendance code that has entered by using the backspace icon when they have entered the wrong numeric character for the attendance code.

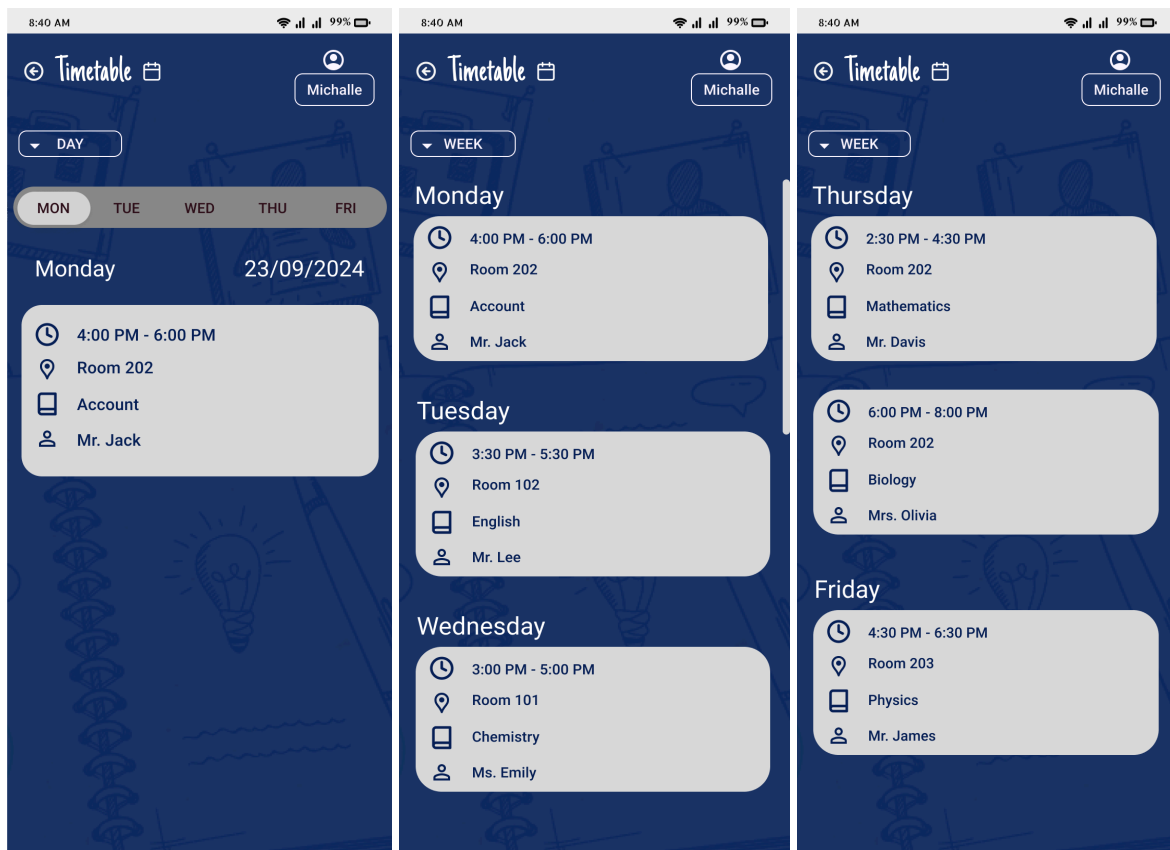
- **Minimal Surprise**

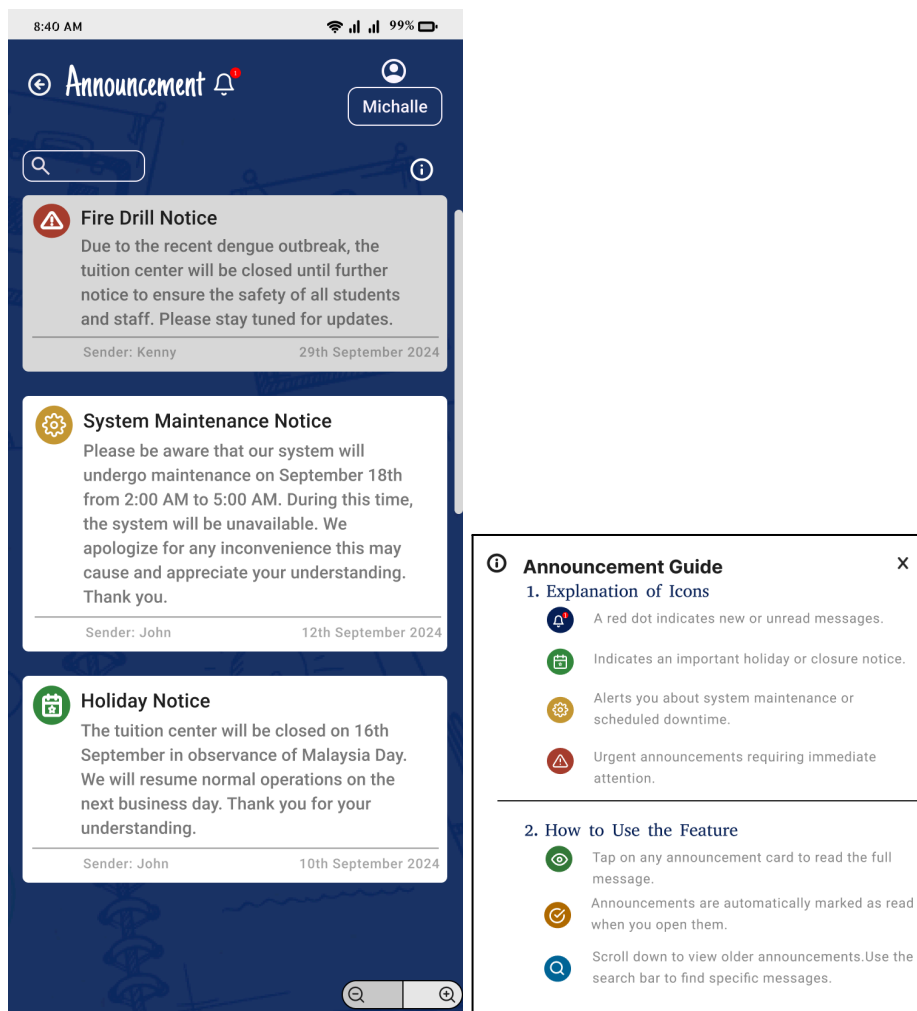
- This concept is also being applied in the application designed as you can see that whenever a user enters the wrong user credentials the application will prompt the useful error message and when the user successfully takes the attendance the green colour check marker was present to indicate that the attendance was successfully taken.

- **User Guidance**

- The application design also provides user guidance in the login section of the application as the sample above shows a sample of user id and password. This will paint a better picture for the user and allow the user to get the prompt faster while making the user feel at ease knowing they are entering the right info to login. Moreover, for the login page, the question mark icon also represents the hint for the users where the password should be at least 8 characters in length including at least a numeric character, an alphabet character and a special character.

11.4 Timetable Management Function & Real Time Announcement Management Function (Gan Khai Li)





• User Familiarity

- The timetable layout follows a familiar day-by-day format, commonly used in educational apps. It is structured in a way that meets user expectations, with a clear sequence of time, subject, and room, making it easy to navigate without confusion. Each timetable view starts with clear day and date labels, using the standard weekly arrangement which is Monday through Friday. The date format (DD/MM/YYYY) is also familiar to most users, ensuring consistency with regional conventions. The class details are presented using familiar icons and labels, similar to traditional timetables. For example, the clock icon represents the class time, the location icon indicates the venue, the book icon shows the subject name, and the person icon identifies the teacher for each session. This consistent use of common symbols and formats helps users intuitively understand and interact with the timetable.
- When applying the principle of user familiarity, users are more likely to feel comfortable and confident using the application because the interface replicates structures and patterns they have encountered in similar educational apps. By adhering to familiar layouts, icons, and formats, the application reduces the learning curve, allowing users to quickly understand and navigate the system. This familiarity

not only enhances user satisfaction but also improves efficiency, making interactions faster and more intuitive.

- **Consistency**

- The timetable layout maintains a consistent colour palette, font style, and box structure for each subject and time slot across all days. This visual uniformity helps users quickly locate key information, such as class times and subjects, without needing to adapt to different layouts. This enhances the user experience by reducing cognitive load. From Monday to Friday, the timetable consistently displays time slots, venue, subjects, and teacher names in the same format. Users can easily anticipate where each piece of information will appear, regardless of the day. The navigation between days (Monday to Friday) keeps the layout and structure identical, ensuring that users don't need to learn a new interface each time they switch days. Each day and time slot follows a consistent hierarchy which is time, followed by subject, then venue, and lastly, the teacher's name. This repetitive pattern simplifies scanning the schedule.
- Consistency helps users quickly grasp how the system functions by ensuring that visual elements and interactive components remain uniform throughout the interface. When layout, colour schemes, and navigation patterns are consistent, users can predict the location and behaviour of features, making it easier and faster for them to interact with the application. This predictability reduces confusion and accelerates the learning process, leading to a more efficient and user-friendly experience.

- **Minimal Surprise**

- The icons used for different announcements are both recognizable and intuitive, reducing confusion. For instance, urgent notices use a red fire icon to indicate their urgency. Each announcement features a clearly labelled title and posting date, similar to other messaging or notification apps, ensuring users are not caught off guard. The layout of the announcements consistently includes the title, content, date, and sender, ensuring that users always know what to expect with each new piece of information. A bell icon at the top of the announcement screen signals new or unread notifications, mirroring the familiar design found in most messaging or social media platforms. This familiar symbol helps users anticipate how they will be alerted to new announcements. Announcements are displayed in a vertical, scrollable list format, a common pattern in apps for reading updates or messages, so users

intuitively understand how to interact with it. There is no unexpected behaviour, such as horizontal scrolling or hidden tabs, which could potentially confuse users.

- Users should not encounter unexpected behaviours or unfamiliar patterns, as this can lead to confusion and frustration. By following the minimal surprise principle, UI design ensures that elements behave in predictable ways such as using familiar icons, consistent layouts, and standard interaction patterns. This predictability helps users quickly understand how to interact with the interface, reducing the likelihood of errors and enhancing the overall user experience.

- **User Diversity**

- The announcement layout includes a clear "Announcement Guide" that explains the meaning of each icon and how to interact with the feature. This guide is useful for both new users who need direction and experienced users who can skip it. Different colours are used to signify levels of importance such as red for urgent, yellow for maintenance and green for holidays. This colour-coding helps users of all experience levels quickly identify key information. Universally recognized icons, such as the bell for notifications, aid users from various backgrounds or experience levels. The announcements incorporate both icons and text, catering to users who prefer visual or text-based information. If a user does not understand an icon, they can refer to the accompanying text for more details, while the text supports those who need additional help with the icons. Additionally, the zoom in/out feature aids users with visual impairments by allowing them to adjust the text size and view more or less content on the screen.
- An UI must accommodate users with varying experience levels, ensuring that both beginners and advanced users can navigate and utilise the system effectively. By applying the user diversity principle, the design provides features such as instructional guides, adaptable text sizes and familiar icons to meet the needs of different users. This approach enhances inclusivity, making the application more accessible and improving overall user satisfaction by addressing a range of preferences and requirements.

11.5 Transport Booking Function & Finance Management Function (Kit Chin Jie Ying)

8:40 AM

99%

Transport Booking

Michalle

Step 1Step 2Step 3Step 4

Step 1: Choosing Number of ways

☐ One way transport

☐ Two way transport

Next

8:40 AM

99%

Transport Booking

Michalle

Step 1Step 2Step 3Step 4

Step 2: Fill up the details

1. Student Name: *
 Fill in your full name

2. Phone Number: *

3. Address: *

4. Day: *

☐ Monday

☐ Tuesday

☐ Wednesday

☐ Thursday

☐ Friday

☐ Saturday

Select which day that need transport only

5. Date: *

For who only require for specific days only

6. Pick up time: *

For those who choose one-way transport put ~-~

7. Return time: *

For those who choose one-way transport put ~-~

* means must fill in

Pricelist
One way Price: RM3 per round
Two way Price: RM5 per round

BackNext

8:40 AM
99%

Transport Booking
Michalle

Step 1
Step 2
Step 3
Step 4

Step 3: Confirm the details

1. Student Name: John Chang Jin Hao
2. Phone Number: 011-11122234
3. Address: 12, Jalan Tiga, 43200, Cheras, Selangor.
4. Day
☒ Monday
☒ Tuesday
☒ Wednesday
☐ Thursday
☒ Friday
☐ Saturday
5. Date: -
6. Pick up time: 8am
7. Return time: 11am

Pricelist
One way Price: RM3 per round
Two way Price: RM5 per round

Back
Next

8:40 AM
99%

Transport Booking
Michalle

Step 1
Step 2
Step 3
Step 4

Step 4: Booking Confirmation



Your booking is confirmed please check your email and the transport timetable will be updated.



- **User Familiarity**

- This application effectively implements consistent design elements, such as the familiar “Next” and “Back” navigation buttons, which are standard in multi-step forms. This ensures that users can intuitively navigate between different steps of the booking process without confusion. Additionally, the information is logically grouped in clear and sequential steps. For example, “Step 1” involves choosing the type of transport, “Step 2” requires filling in personal details, “Step 3” offers a confirmation of the provided information and “Step 4” notifying users that the booking has been successfully completed. This structured approach aligns with users’ mental models of how booking systems typically work, making the process straightforward and easy to follow. Furthermore, the use of a large green checkmark

as a symbol of success or confirmation is a familiar and effective visual cue, ensuring that users clearly understand when the process is complete.

- For the payment system page, this application also utilises familiar logos of well-known payment methods such as PayPal, VISA, Mastercard and Touch N Go, which helps users quickly recognize and understand their available payment options without need for additional explanation information. This form of payment follows a conventional structure, including a field for card details, CVV and expiry date, aligning with users' expectations. This adherence to widely recognized patterns enhances usability, making the process intuitive and straightforward for the user.

- **Recoverability**

- This application includes a “Back” button that allows users to navigate to previous steps if they need to correct any information, ensuring they aren't stuck and can easily recover from mistakes. Moreover, a back button located at the top left corner further reinforces this navigational flexibility, giving users a clear option to go back when needed. Clear rest opportunities are provided in “Step 3”, where users are given a confirmation screen summarising the information they inputted in “Step 2”. This allows users to review and make any necessary corrections before finalising the submission.
- Typically, an error validation for fields like card number and expiry date would prompt the user to correct any mistakes before proceeding.

- **Minimal Surprises**

- This application features a clear step-based progression that enhances the user experience by following a logical sequence. It begins with selecting the type of transport, moving to filling in details, then confirming the provided information, and likely ends with a confirmation message. This predictable flow helps to minimise confusion and keeps the user oriented throughout the process. Each field is labelled ensuring that users know exactly what information is required. This straightforward process eliminated surprises, providing users with clarity and confidence in the actions they have taken.
- The flow of the payment process is straightforward with minimal surprises. Users can see that after entering their payment information, they will proceed to checkout. There are no unexpected steps or extra information requests that would disrupt the process.

- **User Guidance**

- This UI uses step indicators that clearly label each phase of the process, providing users with a clear sense of progression and completion. This ensures that users always know where they are in the process and what remains to be done. In “Step 2”, required fields are marked with an asterisk (*), giving clear guidance on the essential information that must be filled in before proceeding to the next step. Tooltips and explanations are also provided, such as the clarification under the “Day” section, which helps users understand the option they are selecting. After completing the booking, the UI provides a straightforward message instructing users to “check your email”, giving them a clear understanding of what to expect next, thus offering further guidance in the process.
- The UI also provides helpful hints such as “Please enter 16 digits of numbers” for the card number, “Format is MM/YY” for the expiry date, and “3 digit code from your card” for the CVV, guiding users through the process of entering the required information correctly. The “Checkout” button is noticeably displayed and clearly labelled, indicating the next step in the payment process. Additionally, a red note below the form reminds users to verify their information before submitting, further helping to avoid errors.

12.0 Appendix

1. <https://drive.google.com/file/d/1qEo-MAk9ExaKupBLDu9uwHb7IEC429ot/view?usp=sharing>
2. <https://www.figma.com/design/7nxGaBKFK8QHm6UGe3HyVz/Untitled?node-id=0-1&m=dev&t=28IZ1iegZeBmxMm1-1>
3.  Test Case - Cason Soh Kai Sheng
4.  Test Case - Hea Hui Min
5.  test case - Ethel Ng Yi Yan
6.  Test Case - Gan Khai Li
7.  Test case - Kit Chin Jie Ying

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